

Game Concept: CPU² Robots (strongly influenced by the game 'Robot Odyssey' published in 1984 by The Learning Company).

This game mode is specifically designed to be a 2-D programming version of a Robot Odyssey like game. Please refer to the original Robot Odyssey game (or one of its clones, DroidQuest or GATE) for basic reference information.

Similarities and differences between this concept (CPU² Robots) and Robot Odyssey (R.O.):

1. R.O. uses 20x12 screens, with five levels (plus intermediate auxiliary areas) with up to 32 screens per level. R.O. uses screen switching (aka page-flipping). My design would have CPU² Robots using a 256x256 area, where rooms can be defined as custom-sized non-scrollable rooms or rooms that scroll horizontally or vertically. All of the rooms would be fit within the 256x256 area. Extra data would determine how rooms connect or how doors are interacted with.
2. R.O. has four controllable robots. This game will also have four robots along with possible other controllable robots and machines for certain puzzles or points in the story. The CPU² processors would be fully simulated in software. The first three robots will be silver, but will be color-coded with red, green, and blue corners, and the fourth will be black and purple. That robot will be the last robot the player receives and, as the security-enhanced robot, it has a coating giving it resistance to EMP weapons and EM fields. It also has its external grippers enhanced with EMP weapons.
3. R.O. uses logic I/O signals for bumpers, thrusters, radio I/O, and grabber controls. This new design will use flags for input sensing, opcodes for enabling/disabling outputs. Each robot can use one or more processes for simultaneous control of different aspects of the robot.
4. No map parts of R.O. will be copied; the entire CPU² Robots map will be created from scratch, with some similar, but updated concepts and all areas entirely new and distinct. The R.O. city was named Robotropolis; For now, the CPU² Robot city will be called Robostadt as a placeholder for a possible different name. The Robostadt city contains four quadrants. The entirety of this game is played within the first quadrant. The other three quadrants could be potential future map/story/puzzle expansions.
5. R.O. uses logic gates (that are recursively combinable into chips) to control the robots. CPU² Robots use 16x16 or 32x32 programming pages and will have opcode access to all of the standard features of the CPU. Some of the programming areas may be pre-programmed with complex movement programs (like wall following). Some puzzles may require a robot to download and write a new program area for solving the puzzle.
6. R.O. robots were limited in functionality to movement, wall sensing, grabbing and releasing objects, and simple binary signal communications between robots. CPU² Robots will have that functionality along with more complex features such as allowing a robot to "plug" into an interface and performing more complex logic solving or manipulation via programming.
7. R.O. robots have a battery which allow a limited amount of thruster operation before needing to be replaced or recharged. CPU² Robots will have a battery level. This battery level could be used similarly for thruster power or could be used more broadly for CPU cycles.
8. CPU² Robots will have a hardware configuration screen separate from the programming area for various I/O modules such as sensors, shields, alternate movement mechanisms, and filters. Some of these slots will have CPU flags and/or opcodes associated with them for inclusion in programming.
9. Because of the structured design of the robot programming interface, rather than have item storage inside each robot, in this game, the player will have a backpack, accessible by pressing the 'B' key. This backpack can be used to store the robots, as well as any sensor modules, batteries, tokens, keys, and other objects that are not needed at the moment. When the robots are in the backpack, an icon will appear at the bottom right of the screen for each robot. The player can click on the icon to access the robot or double-click on the icon to remove the robot from the backpack. The player can access a robot interface by coming in contact with the robot and pressing the 'E' key. Once inside the interface, the player can use the 'P' key for the periscope, the 'B' key for the backpack, and the 'C' key for the robot configuration screen.
10. CPU² Robots will have a 'Robot Display' register which can receive up to eight characters. When the register is written to, on the map, the robot will briefly have a speech bubble containing the characters. The character set is a combination of ASCII and a robot-gibberish character set. Robots that are direct in contact with one another can send a display message from one robot to the other. When a message is successfully sent, the receiving robot will repeat the message in a speech bubble. The robot display might have a purposeful lag or cooldown time associated with using it to prevent it from becoming a replacement for radio communications.
11. Power cell chargers are used in R.O. to recharge cells. Some of the rechargers are locked behind simple puzzles. In CPU² Robots, battery cells are expended and disappear once used inside a robot. If the player has a robot run out of power, a variety of mundane tasks and simple puzzles are available for the player to earn battery cells. The player should never face a situation where game progression ends because the robots have had their power depleted. There might also be other non-puzzle tasks for the robots to perform to earn money.
12. The four robots the player programs in CPU² Robots will not be simple machines in this game like they are in R.O. They will have NPC dialogue and be featured in most parts of the overall story. Their primary dialogue purpose (outside of the story arc) will be to explain new environments, orient the player as needed, and remind the player as to what steps need to be taken to move forward with the game progress; crucially, this includes using plausible NPC dialogue to better explain a puzzle in a way that couldn't be done passively. This will allow more interesting and complex puzzles to be implemented in the game in a way that isn't game-breaking or overly confusing for the player. Other aspects of NPC dialogue could be explored; the different robots might have different personalities and offer slightly different takes on how they perceive and report things. Some NPC dialogue could be presented as an optional drill-down dialogue tree that allows the player to stop receiving dialogue about a specific topic when they are satisfied with what they have received so far. Possible idea: Some of the story arc discussions might be presented as cut scenes designed to look like a static comic book page with panels. Sequential panels and word balloons will appear with dynamic text filling in to simulate talking by the characters. A rough example I am thinking about is the original 2001 Max Payne video game, except with word balloons appearing and filling in as the conversation takes place. If a cut scene is not appropriate, word balloons might appear from the individual robots either inside or outside the backpack. One rule would be that a cut scene would never appear if any of the robots are running a program, but individual dialogue might still show up. Verbal dialogue/voice acting will not be used in this game.
13. The robots in R.O. had cardinal direction based (north/south/east/west) directional thrusters, wall bumpers, and other sensor objects. The CPU² Robots can switch between compass (cardinal direction) mode and relative mode, which renames thruster, bumpers, sensors, etc. to front/back/left/right, and adds rotate left and right opcodes for the robot movement. The compass direction mode is locked behind needing a compass module. In the relative thruster mode, the rear thruster is stronger and moves the robot at double normal speed, and the left, right, and front thrusters push at regular speed.
14. Near the end of the main story arc, the player will gain the ability to program other random robots to complete a set of maze rooms in a challenge. The random robots will feature various standard, unusual, or downright exotic configurations resulting in program challenges of various difficulty levels. Other optional programming challenges will also be featured during the game.
15. Saved games for CPU² Robots will be slightly different than other games. The portion of the save that records where the player is, what they have accomplished, what items they have, etc. will be stored conventionally, but their grid programming code saves will be stored as a per-page version history for each robot. If the player loads an older save, the pointers for which code pages are active will be moved back to the pages that were active when the game save took place, but the player is free to slide the pointer for an individual page forward or backward to whatever version of that page they want. The player can also click the "save" button on the programming screen to save that code page in the version history whenever they want. This is a way to guarantee that no player grid code is ever accidentally lost, even if the player loads an old game save. The only removal of this data would occur if the player deletes all of the game saves for that game play-through, then the whole save area for that play-through will be deleted. On the displayed code page timeline, manual code page saves and game saves will be shown most prominently, auto-saves will be less prominent, and automatic session saves (whenever the player exits the modified code page window) will show up as dots. The code page saves will be stored as efficiently as possible to minimize disk space usage.

CPU adjustments and other information for this game:

1. CPU will be 32-bit, with a 64-bit macro area.
2. A configurable page mode will be added to implement configurable multi-size program pages. A 'page' register segment will be added to the MAR and PC registers, as well as the mini-stacks for the LOOP and JSR subsystems.
3. The programming will utilize most of the current standard opcodes 0x00 thru 0xBF. Obvious additional opcodes used for robotic functionality can be defined in the 0xE0-0xFF areas. A non-shared map memory (one 256x256 area per CPU) will be used for definable page areas for programming. The user can select which specific program area is default for each robot, allowing storage and reuse of programs. The robot will also have the programmatic ability to switch execution between pages. The current program counter consists of an 8-bit X coordinate, an 8-bit Y coordinate, and a 2-3 bit direction. The program counter for this new mode will be variable, depending on the size of page being accessed, with a total of 16 X/Y/page bits plus 3 direction bits.
5. New flags will be defined for bumpers, gripper sensing, interface, radio, and remote control. Current flag subsystem will have standard CPU flags 0-31 remaining how they are configured, and flags 32-55 configured for CPU² Robots. Flags 56-63 are unused.
6. Each robot can utilize multiple processes. The initial process will start in the lower left corner (relative coordinate 0,0) of the page assigned to the robot. Processes can be forked (duplicated) to become dedicated to thrusters and movement, bumper processing, or radio communications.
7. Unlike the CPU clock setup planned for the regular CPU² Grid Wars game mode (which runs at a dynamic speed to optimize viewability of watching the game combat), the CPU² Robots game mode will have the CPU run at a high rate dependent on the granularity of movement of the robot, such as 64 full process execution cycles of all processes, for each pixel that the robot moves. This is necessary to prevent the robot from missing or lagging on an event such as a wall collision or radio communications. This speed will need to be optimized for the game.
8. Certain pre-defined programs might be loadable into the robots program area to act in a similar was as pre-defined chips in R.O. This could also be extended to allow a robot to simultaneously run processes on multiple areas defined for specific functions such as complex movement, communications, and complex object manipulation. User pages will be numbered starting at page 0, and game-supplied pages will be identified with letters A-Z. Pages will have a 16-character title, with numeric pages editable by the player and letter-coded pages hardcoded into the game. In-game program cards can unlock specific programs. These cards are available from some puzzles or NPC robots in the game. There will be a special spot in the backpack for storing these cards. When starting to use a new page, space will be automatically allocated from the page memory corresponding to the size of page requested. A form of auto-layout will be used to optimize space in the overall program area map.
9. The thruster on/off opcodes are simply setting or clearing the 'Engage Thruster' flags, which would be what actually controls the thrusters. Each robot individually has a radio communications flag (1-bit register) that it can toggle (to send a high or low signal), but when reading the flag, it effectively is a logical OR of the four robots radio flag registers. A robot will not be able to distinguish whether it is sending a radio high signal or if another robot is, so programs will need to be written to ensure the radio flag is cleared when not sending in order to receive properly. The read-only gripper and interface flags are set to true when the gripper has grabbed something or the interface has been connected, respectively.
10. The interface register could be padded byte-wide or 32-bit, or might possibly require multiple reads to obtain all of the data depending on the puzzle design. Some puzzles might require both transmitting and receiving data from an interface.
11. For some of the more advanced puzzles, newer machines need to be programmed in similar ways as a robot, except they would use a hexagonal or triangle program grid instead of a square grid. The drone might use a hexagonal grid instead of a square one, to indicate a more advanced or exotic technology. Triangle grid robots and machines would only be used for the most difficult puzzles and obstacles, since the programming would be so challenging. The triangle grid is likely limited to 16x8 or 32x16 instead of the usual 16x16/32x32 due to its layout.

CPU² Robots Game Subsystem	
THR,N,OFF	Disengage North Thruster
THR,N,ON	Engage North Thruster
THR,E,OFF	Disengage East Thruster
THR,E,ON	Engage East Thruster
THR,S,OFF	Disengage South Thruster
THR,S,ON	Engage South Thruster
THR,W,OFF	Disengage West Thruster
THR,W,ON	Engage West Thruster
GRIP,OFF	Disengage Gripper
GRIP,ON	Engage Gripper
IFC,OFF	Disengage Interface Connection
IFC,ON	Engage Interface Connection
THR,A,OFF	Disengage All Thrusters
IDLY**	Instruction Cycle Delay ## x 1 Cycles
TDLY**	Tick Delay ## x 1 Game Ticks or Frames
RREG,IFC	Read from Interface
WREG,IFC	Write to Interface
RREG,RD	Read from Robot Display Register
WREG,RD	Write to Robot Display Register
RREG,MC	Read from Memory Card Slot
WREG,MC	Write to Memory Card Slot
DMODE,C	Compass directions, North/South/East/West
DMODE,R	Relative directions, Front/Back/Left/Right
MCRST	Reset memory card interface.
RROT,R**	In Relative mode, rotate right ## degrees
RROT,L**	In Relative mode, rotate left ## degrees

** Two-byte opcode, second byte is data value.
*** Three-byte opcode, second and third bytes are data values.

CPU² Robots Alternate Direction Mode	
THR,F,OFF	Disengage Forward Thruster
THR,F,ON	Engage Forward Thruster
THR,R,OFF	Disengage Right Thruster
THR,R,ON	Egage Right Thruster
THR,B,OFF	Disengage Back Thruster
THR,B,ON	Engage Back Thruster
THR,L,OFF	Disengage Left Thruster
THR,L,ON	Engage Left Thruster

Extended Sensor 2 Robot Flag IDs:			
Numeric	Mnemonic	FS1 Bit	Description
40	NES	b8	Northeast/RightFront Sensor 2
41	SES	b9	Southeast/RightBack Sensor 2
42	SWS	b10	Southwest/LeftBack Sensor 2
43	NWS	b11	Northwest/LeftFront Sensor 2

Extended Controller 2 Robot Flag IDs:			
Numeric	Mnemonic	FS1 Bit	Description
56	NEC	b24	Northeast/RightFront Controller 2
57	SEC	b25	Southeast/RightBack Controller 2
58	SWC	b26	Southwest/LeftBack Controller 2
59	NWC	b27	Northwest/LeftFront Controller 2

	0xE_	0xF_
0x_0	THR,N,OFF	RREG,IFC
0x_1	THR,N,ON	WREG,IFC
0x_2	THR,E,OFF	RREG,RD
0x_3	THR,E,ON	WREG,RD
0x_4	THR,S,OFF	RREG,MC
0x_5	THR,S,ON	WREG,MC
0x_6	THR,W,OFF	DMODE,C
0x_7	THR,W,ON	DMODE,R
0x_8	GRIP,OFF	
0x_9	GRIP,ON	
0x_A	IFC,OFF	
0x_B	IFC,ON	
0x_C	THR,A,OFF	MCRST
0x_D		
0x_E	IDLY**	DROT,R**
0x_F	TDLY**	DROT,L**

Robot Flag IDs:			
Numeric	Mnemonic	FS1 Bit	Description
32	NB	b0	North/Front Sensor 1 (bumper)
33	EB	b1	East/Right Sensor 1 (bumper)
34	SB	b2	South/Back Sensor 1 (bumper)
35	WB	b3	West/Left Sensor 1 (bumper)
36	NS	b4	North/Front Sensor 2
37	ES	b5	East/Right Sensor 2
38	SS	b6	South/Back Sensor 2
39	WS	b7	West/Left Sensor 2
40	-	b8	Reserved
41	-	b9	Reserved
42	-	b10	Reserved
43	-	b11	Reserved
44	GA	b12	Gripper Active
45	IA	b13	Interface Active
46	AS	b14	Area Sensor
47	CS	b15	Contact Sensor
48	NT	b16	North/Front controller 1 (thruster)
49	ET	b17	East/Right controller 1 (thruster)
50	ST	b18	South/Back controller 1 (thruster)
51	WT	b19	West/Left controller 1 (thruster)
52	NC	b20	North/Front controller 2
53	EC	b21	East/Right controller 2
54	SC	b22	South/Back controller 2
55	WC	b23	West/Left controller 2
56	-	b24	Reserved
57	-	b25	Reserved
58	-	b26	Reserved
59	-	b27	Reserved
60	RX	b28	Radio Transceiver
61	HR	b29	Hover Reversal
62	MC	b30	Memory Card Sensor
63	-	b31	Unused

* Memory Card:
The memory card functionality is made up of three pointers, a read index pointer, a write index pointer, and a end-of-file pointer. When the memory card register is initially written to, the 32-bit data is written to the location specified by the write index pointer (zero), its write index pointer and end-of-file pointer are then incremented by one, to the next 32-bit word. The program can continue to write additional values this way, incrementing the write and EOF pointers until a memory card read takes place or the program is terminated.
When a memory read takes place after a write, the write index pointer is reset to zero, but the end-of-file pointer is left alone. The first 32-bit word is sent to the memory card register and the read index pointer is incremented by one to the next 32-bit value. If an attempt to read beyond the end-of-file pointer location occurs, the CPU should set the 'F' fault flag, reset the read index pointer to zero, and return a zero in the memory card register. If the read is successful, the 'F' fault flag should be cleared. If a write takes place after a read, the read index pointer and end-of-file pointers should be reset to zero, the value should be written to the first location, and the write index and end-of-file pointers should be incremented.
If the MCRST opcode is executed, the read index and write index pointers should be reset to zero, but the end-of-file pointer will be left alone. If a memory card is removed on the robot configuration panel, the end-of-file pointer should be stored with the removed memory card's contents. If a memory card read or write is attempted without a memory card present in the slot, the 'F' fault flag should be set. The 'MC' flag can be checked for the insertion status of the memory card (false=empty slot, true=card present). Other options available for the memory card interface could be to limit it to 8-bit or make it switchable between 8-bit and 32-bit.

- Critical story elements to address:
1. How did the human, Ash, get into the robot city? And why?
 2. What is the reasoning (and backstory) behind the need for Ash to help via robot grid programming.
 3. What is the desired outcome for the robots that Ash can help with?

- Other minor story elements:
1. Humans in the robot city: how they survive and why they are allowed to be there.
 2. Feasibility of a human being able to program and modify robots.
 3. Explanation for how a human can carry multiple robots.

Possible Ash intro story:

Having been granted her uncle's house from her deceased uncles estate, she was also given her uncles archeological research papers contained in the house. Ash tried to figure out for herself her uncle's research that he was always so unwilling to describe in anything more than extremely vague and unsatisfying language. Her uncle was always quite wealthy, yet was never associated with any museum or university and never published any research papers. Where did his money come from? Illegal antiquities trade? The only clue she ever saw was a one kilogram gold bar with a strange symbol on it that fell out of one of his pockets one day as she was visiting – he quickly scooped it up and insisted that she must have imagined it. Digging through the nonsensical notes with strange symbols scribbled in pencil in the margins, Ash ultimately found in one of the stacks a map with enough landmarks to allow her to identify the geographic location the map corresponded to. The map was marked with two teardrop symbols, one with a letter C inside it and the other with a letter U inside it. She suspected that the letter C was short for "cave", and she was determined to go take a look. Having recently quit her veterinarian assistant job, Ash called her mother to ask her to keep an eye on her animals. After dropping a spare key off at her mothers house, she drove out to explore the cave and spend a week or so camping and relaxing.

Ash followed the smudged and faded map as best as she could, with roads becoming rougher and more impassable as time went on, some almost appearing to have been purposely altered to be as inhospitable as possible. As the sunlight was beginning to fade, she finally arrived at where the cave should be. Ash looked around and found a fallen tree with the teardrop 'C' symbol carved into its bark; she knew she was in the right area, although she was now not so sure about what the symbol meant. As she continued past the tree, there was nothing but razor-sharp blackberry bushes covering the side of a hill. Using leather gloves and a machete to painstakingly chop a path through the bushes, she eventually came across a heavy wooden panel of what looked like teak partially covered by dirt and firmly embedded in the 30-degree slope of the hill. In the center of the panel, she could faintly see a stenciled silhouette of the same teardrop symbol, faded by years of constant sun exposure. At its base, the panel had a pair of rusty strap hinges connected to what appeared to be a buried and reinforced railroad tie. A gentle rain had started falling and a chilly breeze was in the air. She used her gloves to dig the dirt away, becoming more and more concerned about the weather. After about half an hour of digging, she managed to pry up the upper edge of the hinged panel and expose a small cave opening with a rope attached to the panels upper corner and a rope ladder leading diagonally down into the deep, dusty darkness. As Ash climbed down into the hole, she realized that the cave she was entering was a reinforced concrete enclosure with steel beams. She reached the bottom and started looking around with a flashlight and found what looked like an old mining lift with a sign next to it. The sign had an arrow pointing up toward the ceiling and stated, "Extremely Important: Use the rope to keep the door closed." Without really thinking it through, Ash used her full body weight to pull down on the rope dangling from the ceiling next to the rope ladder, and the outside wooden panel she had struggled to pry open swung closed with a solid thud. Realizing she was now likely stuck in the underground room, she carefully examined the lift and decided to explore further. She stepped into the lift and tried to pull the gate closed. After a bit of fiddling with the lift, a dim amber light lit up on the wall indicating to Ash that the lift had power. With an uneasy sigh, she tugged on a large metal lever to see if the lift would start descending. When it suddenly started downward, she was both excited and terrified. She was finally going to see her uncle's secret archeological looting site. The lift continued to descend, when it suddenly started repeatedly shuddering and falling, screeching and slowing down, then shuddering and falling again. This repeated again, and again. She was knocked out when the lift finally hit the bottom of the shaft.

As Ash slowly regained consciousness and while still in a daze, she opened her eyes to find a large blurry figure cloaked in black scuffling with a gray figure on what looked like a brightly lit street corner. After the battle, the gray figure dropped something large and metallic as they were being dragged away. Once finally able to see clearly and stand up, Ash marveled at how the area was made to look like an old inner-city neighborhood, except with a 15 meter high smooth rock ceiling – it looked like a bunch of residential buildings had been teleported into the carved-out tunnel. She looked over at how the lift was now disconnected and completely mangled. Above its entrance was the stylized teardrop symbol, its point angled diagonally down and to the right, with a circle in the middle. After a faint sound caught her attention, Ash walked over to examine the large silver box dropped by the gray figure. The box appeared to be a slightly trapezoidal-shaped aluminum box and had what looked like backpack straps on the largest surface. She flipped it over to see four large square cavities, three of which looked empty. Suddenly, a tiny distorted voice told her to take this "carrier" and quickly go to the first door on the left. Seeing what appeared to be another black figure in the distance coming towards her, she quickly grabbed the metal box and glanced down the street. She was finally paying attention this time, trying to make sense of it all: she was standing in the middle of a cobblestone street, slowly arcing to the left in the distance, with both sides lined with various stooped brownstone apartment buildings seeming to go on and on. Lit by street lamps, apartment windows, and dim gray-blue ambient light sources shining up onto the smooth rock ceiling, Ash really felt like she was walking around in an actual city at night. The lack of cars on the street made the whole place feel abandoned and the lack of sidewalks gave it a European feel. There were strange yellow and black striped partitions set up to block the street at certain points.

Once Ash went up the steps and into the building, she went down a short hallway and opened the first door to find what appeared to be a nice apartment with a bedroom, kitchen, and bathroom connected to a large living room. She stepped into the room and pulled the door closed. The voice told her to place the carrier flat on the floor and step back. Once she complied, a small green and silver metal cube floated out of one of four square openings on the carrier and grew rapidly to a size slightly larger than a meter, floating about a quarter meter off of the ground. After a few seconds, a cylinder with a round plate on it raised up out of the top of the cube and spun around to reveal a simple robotic-looking face wearing a green hat. A pair of arms shifted out from cavities in the cubes sides and unfolded. Suddenly the face spoke to Ash and introduced itself as "Viz." Ash stumbled back and stood in a stunned silence, unable to say a word at what she just witnessed. Viz, sensing that this may be this human's first visit, proceeded to give her a brief history of the robot city and the rules for humans visiting the city. After rambling on and on, Viz suddenly realized how fatigued Ash had become. Viz asks Ash to plug it into a socket on the wall and then tells Ash to get some sleep; they can continue talking in the morning. Ash went into the bedroom and cautiously laid down on the bed as Viz started their chassis diagnostic routine and shut itself down for the night.

Viz will be the first of many robots that Ash will befriend in the robot city of Robostadt.

The cargo bots (background for story):

The red, green, and blue compact cargo bots (Sol, Viz, and Mel) work as a team to transfer robot-building materials between sites. During normal work days, the three robots spend all their time in the research sector moving materials between buildings. Once a week, the they have a special time-limited task to transfer materials between the industrial, commercial, and research sectors. During these operations, they work as specialized transporters: One transfers raw materials from Refining & Reclamation to Robolithography for chip making. Another one transfers new CPU modules and other chips from Nanoscale Assembly to Base Logic for low-level ROM programming and chassis setup. The third transfers modular robot chassis cores and certified CPU modules from the 'Bot Proving Grounds to Automaton Training for high-level training and personality setup. Sometimes one of them swaps out to deliver silicon boules from the Silicon Crucible to Robolithography.

For the weekly operations, a fourth compact cargo bot (Lex) is brought in to monitor and provide limited security for the three robots and to transport high-security or prototype materials. Lex (black with purple trim) has an EMP-resistant ferro-ceramic coating and EMP weapons installed above its grippers, but is otherwise an identical model as the other three. Lex was also issued a modular carrier strapped into its cargo compartment used to recharge and protect the other cargo bots in an emergency. When it is not on one of these weekly transport missions, Lex is providing secure chain-of-custody cargo delivery for some of the research areas.

The cargo bot incident:

The four cargo bots were tasked with a cross-zone delivery of a double order. Due to the recent sector shutdown and sparse crowds, the mission plan directed the robots to load up and deliver two loads instead of the usual single load. The red bot, Sol, was assigned to deliver gold and aluminum ingots, the blue bot, Mel, was assigned to silicon boule delivery, the green bot, Viz, was assigned to CPU module delivery, and Lex was delivering prototype mega-CPU modules. The first delivery was uneventful, with all of the bots delivering their loads successfully.

While the robots were going back for their second loads, Viz suddenly found itself surrounded by several unknown robots in the midtown district. The thugs grabbed Viz and overpowered its thrusters, pushing it into a side alleyway. Viz sent a radio alert to Lex, who was at the other end of Midtown. Lex, after directing Sol and Mel to hide wherever they could, spent time searching the district for Viz. Using their private radio channel, Viz was able to give Lex directions to where it was cornered. Lex surprised the thugs and pointed its EMP weapons at them. The thugs quickly fled. Lex briefly went after them, but ended up returning to protect Viz from being attacked again.

After Lex directed Viz to shrink down, Viz shrunk from a roughly 1.2m cube down to an 18cm cube. Lex opened its cargo compartment and the mini-Viz hovered up into the compartment and into an opening on the carrier, magnetically attaching itself in place. Just before Lex closed its cargo panel, it and the other cargo bots received a strange digital radio signal that rendered them unable to move or use their other external peripherals. The criminal bot saw Lex suddenly stop moving and gingerly walked up to its open cargo area. At that moment, the bot realized that Lex had an active homing beacon pinging away, so it grabbed the carrier, locked Lex in a storage unit and fled. Lex, still in radio contact with the other three cargo bots, confirmed with them that they were all unable to move. After attempting to see if a reboot fixed its chassis problem, Lex had no other choice but to direct Sol and Mel to power down to conserve power, and direct Viz to stay running in low power mode in order to track the criminals movement. Lex then shut itself down to also conserve power.

Because the cargo bots missed their mandatory 3-hour check-in after the weekly transportation task, their special express transport passes had expired and they are now stuck on whatever level they are on. Ultimately, after an incident that disabled the criminal bot, Ash obtained the backpack and Viz.

The CPU-chassis interfaces are still disabled on the four cargo bots, so they cannot move on their own – they will have to rely on Ash to either carry them in the backpack or learn how to program them for movement using their grid interface. Sol and Mel had successfully managed to hide before being affected by the radio signal – Mel is in the commercial district and Sol is in the Industrial district. Lex is locked away somewhere in Midtown.

Note: All story and lore ideas presented on these pages are just my ideas used to ultimately flesh out puzzle concepts. All aspects of the story and lore could change or be discarded or taken and fit within the game direction.

Possible cargo bot story timeline (story spoilers):

1. Lex gets its chassis upgraded with an anti-electromagnetic coating. Because of the added EMP weapons being non-standard, Lex gets recertified in the Bot Proving Grounds. Unbeknownst to Lex, the shielding has blocked their harmonic identifier from being recorded so the recertification ends up being invalidated later on.
2. As the one who handles the finances, Lex submits the cargo bots' annual condominium lease payment to the government for their apartment.
3. The government announces a nuclear reactor downtime is needed to maintain the steam tubines. This will affect all large power users (large industrial, commercial, and research facilities.) Large facility operation will be shutdown or reduced due to insufficient power capacity for the next two to three weeks, but shops, banks, and other smaller businesses and less power-hungry operations such as shipping and delivery may remain active. Inter- and intra-zone paths will be closed or restricted to reduce crime risks. Employees of closed facilities should stay home or stay within their residential zone to reduce congestion. Robots or humans with special passes will be allowed inter-sector access.
4. The Cargo bot incident story occurs. The cargo bots lose their movement functionality and are at various locations around lower Quadrant 1.
 - 4a. The four cargo bots were tasked with a double-delivery during the citywide shutdown.
 - 4b. The first delivery was uneventful.
 - 4c. Thugs attempt to kidnap Viz in Midtown, and end up grabbing Viz and locking Lex away in a storage room. Sol and Mel successfully hide.
 - 4d. The four cargo bots ability to control their chassis has been remotely disabled.
5. Ash's intro story occurs. As Ash arrives on the old lift, she witnesses a black sentry arresting another figure. That other figure is the 'criminal' bot that targeted Viz and Lex in Midtown. Ash rescues the green robot, Viz, in Lower Robostadt and is directed into an unused human apartment to get some sleep.
 - 5a. Ash gets a map from her deceased uncles archeological research documentation.
 - 5b. Ash decides to go to what she thinks is her uncle's secret cave dig site.
 - 5c. After some rough terrain and unpleasant surface vegetation, Ash finally arrived at and uncovered the cave opening, only to discover it is some type of mine.
 - 5d. Ash takes the lift down into the mine, but the lift fails and ultimately breaks looks from its cable and falls. Ash is knocked out.
 - 5e. Ash sees a scuffle between two figures, then sees one drag the other away. Ash investigates an item one of them dropped.
 - 5f. Ash realizes she is in some sort of underground neighborhood made up of brick buildings.
 - 5g. A small voice tells Ash to grab the dropped item and head for the first building. Ash complies and enters an apartment.
 - 5h. The item Ash grabbed was some sort of aluminum backpack. She places it on the floor and a small cube hovers out, enlarging to a large cubic-shaped robot.
 - 5i. After some brief discussion, the robot, Viz, gives Ash an introductory chat before telling Ash to get some sleep.
6. The next morning, Viz explains the strange situation with their chassis and spends some time instructing Ash about the shrink capability and the carrier/backpack unit, how the grid programming is accomplished, and various other things she will need to know to get around Robostadt.
7. Ash is informed that the first building on either side of the old lift are human apartments (marked with a red human silhouette sign) and the second building on the left is the human services building (marked with a red heart sign) containing the library, medical clinic, food storehouse, exercise area, and theater room. Ash goes to the library and starts learning the basics of grid programming. She queries one of the robots about human food and she is delighted to find lots of robots growing their own gardens, identified by having a plant in the front apartment window. She loads up on fruits and vegetables and goes back to the apartment to eat.
8. Once Ash feels ready to begin the journey, she loads Viz up into the carrier. Viz informs Ash that the industrial district is split up into many zones that have been sealed apart via their fire doors, and that they would need to work through industrial facilities to cross from zone to zone to get to Sol. The first industrial facility they enter is the Bot Proving Grounds.
9. Ash meets the Bot Proving Grounds evaluator and is told that only unofficial tests can occur because the official trainer is stuck in another district. After quite a few tries, Ash successfully programs Viz to complete the course and enters the next industrial zone.
10. Ash and Viz work their way through the Chassis Forge and Refining & Reclamation facilities in the industrial district to rescue Sol.
11. They reach a government field office at the far end of the industrial zone. A government robot looks up the information and tells them that the cargo bots received an autonomous summons to court for an unpaid lease bill which has disabled their ability control their chassis. Because their cargo chassis did not possess the correct functionality to travel to the courtroom they were summoned to, they are now stuck in this disabled state. The government bot tells them that they would have to talk to their bank to find out why their payment didn't go through. Only the judge or a mainframe can override the summons. Paying the bill should also get the summons automatically cancelled.
12. They continue on to Midtown. After exploration and continued training at the Midtown library, Ash and the bots continue into the commercial sector.
13. Ash, Viz, and Sol cross through the commercial sector to rescue the blue robot, Mel. The four of them take the subway to Upper Robostadt.
14. Ash and the bots reach the Robobank branch in Upper Robostadt. The bank manager indicated that Lex's payment never went through because Lex's robot status is currently decertified and Lex needs to get re-certified by the government to unfreeze its bank account. That can only occur with a special authorization from the central mainframe in the government district.
15. In Upper Robostadt, Viz recognizes the 'criminal' bot from Lower Robostadt and Ash confronts them. They discover that this robot is a former private debt collector that was arrested for theft and recently released from jail on bond. It tells the cargo bots that it was going to collect the missing lease payment when it panicked and fled when they saw Lex's homing signal and thought it was signalling a black sentry. In exchange for keeping quiet about the incident down in Midtown, the former debt collector bot gives Ash a key, a lower-Robostadt express pass and directions to the room that Lex is locked in.
16. Ash goes and rescues Lex, and tells Lex about the autonomous summons. Lex indicated that it had a suspicion that when it was at work, its special coating blocked coded transmissions, so any notices of non-payment of bills, government messages, etc. would have all been missed if they were sent during working hours. Lex indicated that they would talk to research about adding in a special circuit to retransmit the coded messages through the anti-electromagnetic coating similar to how their identity harmonics can be rebroadcast outward through the main antenna. Either that, or for research to get another unshielded compact cargo chassis for Lex to use during normal working conditions and reserve the shielded chassis for special occasions. The other bots expressed how strange it would be to see Lex return to using a regular purple and silver cargo bot chassis.
17. Ash returns to Upper Robostadt to continue her programming training sessions and visiting robot gardening hobbyists to sample fresh fruits and vegetables and offer gardening advice.

(Timeline continued on next frame)

(Timeline, continued)

18. Ash and the four bots enter the research sector and work through the Logic foundry, Robolithography, and Nanoscale assembly facilities.
19. A friendly research robot who recognized the bots gives them useful information: robot recertification occurs at the Bot Proving Grounds, but that the training bot is unable to leave their residential zone. Even with the authorization, they would have to wait for the subway upgrade to be completed in a couple of weeks.
20. Ash continues up into the Skyline district. Ash experiences more and more advanced programming training and very exotic food from local gardens. The human apartment in the Skyline district is the biggest and most luxurious place she has ever seen, and watching the supercomputer operation in the evening is more amazing than any fireworks display she's ever seen. In her spare time, she tries out some of the crazy puzzles created by the gen 4 androids.
21. Ash and the bots enter the Robogov district and cross several government facilities to reach the Mainframe Collective.
22. Ash and the cargo bots finally reach the mainframe console in the Mainframe Collective, and discuss Lex's decertification with the mainframes. The mainframes realize a mistake has been made somewhere* and they issue Lex a recertification authorization, but they also mention the lack of the training bot. Trying to find a quick solution to this issue, several of the mainframes examine the four cargo bots grid programming logs and are impressed by the improvised programming that the human has created. They grant Ash a temporary 'official trainer' permit for the Bot Proving Grounds along with an express pass for quick access back down to the lower sectors. They code both into a harmonics band and give the band to Ash to wear.
23. Ash travels back down to the Industrial district and shows the evaluator bot her temporary trainer permit. The evaluator examines it in astonishment – no human or robot has ever received a "temporary" permit for this. It then lays out the recertification rules: the robot must go through the whole course alone without a single fault or reset within the restrictive time limit, and that even a single failure will mean having the restart the whole course over again. Ash, as the trainer, is allowed as many test runs as necessary to program Lex for the recertification, and is allowed to reposition Lex between rooms and select the appropriate program at the beginning of each room. This time, Lex makes sure its harmonics identity is being broadcast while it is taking the test, and makes sure Ash is wearing her harmonics band.
24. After Lex successfully completes the course, the evaluator bot submits the recertification to the government, warning that approval will take a few weeks. Because of the number of bots waiting to be certified, the evaluator bot urges Ash to certify the other bots waiting in line** so that they don't have to wait weeks in line for the return of the regular trainer bot. It indicates that a few of the bots new employers might be willing to offer rewards for quick certification for the bots. After certifying 10 robots, the evaluator tells Ash that RoboExpand in the commercial district is interested in seeing how well she works as a tester for their data processing certifications.
25. (optional) Ash goes to RoboExpand and certifies three robots waiting in line. After certifying the robots, the RoboExpand evaluator apologizes for the lack of robots and tells Ash that Automaton Training in the research sector is always looking for new trainers for their advanced robot certifications if she wants a bigger challenge.
26. (optional) Ash goes to Automaton Training and certifies random robots at that facility, in a similar fashion as the Bot Proving Grounds. After certifying 10 robots, the Automaton Training evaluator congratulates Ash and tells her that the Mainframe of Robotic Health & Growth has been monitoring the certifications and has granted Ash a permanent trainer license.
27. Ash completes any extended story arcs and puzzles.
28. When Ash has finished all the missions and wants to leave, she and the bots go up to the new surface lift in the government district. The room has a bank of power/terminal sockets along one wall, and Ash plugs each robot into a socket to await their getting their chassis functionality back. Ash still has the old map from her uncle, now with added instructions scribbled on the back about how to hike back to her truck. Everyone says goodbye as Ash drops the backpack and steps into the lift. Ash tells the bots she promises to return for a visit. The lift rises up until Ash has disappeared.
29. Just after Ash leaves, Lex receives notification that its recertification has been accepted, and then receives a notice that the lease payment has been received. Suddenly the bots are able to move again. Lex admits that it has been able to move on its own since right after its recertification but that it felt sorry for its other three friends and decided to just play along. The bots move off-screen, then Lex reappears and goes and grabs the dropped backpack, then disappears again.

* Possible optional story arc branch: Ash is determined to find out why Lex was incorrectly decertified. In doing so, the bots uncover the alien origins of the robot city.

** (optional) Robots with varying configurations will be presented, most will require brand new programming and may be exotic robots with strange or limited configurations. Five robots will be in the line for Ash to choose. Each one successfully certified will cause a new robot to enter the line. Ash can continue to certify bots until she has five robots that she is not able to certify, either because of her skill level or because the robots have unworkable configurations. The first five robots in line will always be the same, each additional one will be random.

Ultimate lessons learned from this story:

1. The four cargo bots are only certified in their compact cargo chassis, which normally, automatically covers their simpler bipedal chassis use. Because Lex lost its cargo chassis certification, it was no longer certified in anything (officially decertified), which partly caused this whole mess. All the bots should get certified in their bipedal chassis as a backup.
2. All four bots were paying Lex their portions of the lease and depending on Lex to send the bill. They need to rework that so that there is some redundancy available to make the lease payment.
3. Lex needs to ensure that it is receiving all the required transmissions when it is using its hardened cargo chassis. It should also request an additional non-hardened chassis to use for normal circumstances. This will also extend the service life of the hardened chassis.
4. At the Bot Proving Grounds, if the evaluator cannot detect the harmonics of the bot under test, it needs to speak up immediately about it rather than quietly invalidating the test after the bot under test has left.

Note: All story and lore ideas presented on these pages are just my ideas used to ultimately flesh out puzzle concepts. All aspects of the story and lore could change or be discarded in favor of a different fit within the game direction.

Possible “part 2” story:

Ash had just gotten home from the bank, talking to the bank manager about mortgaging her house and getting her late uncle's former residential property rezoned and upgraded into a small animal hospital she wanted to start. Having received her doctorate degree in veterinary medicine a couple of years ago, she was firmly on her way to doing what she always wanted to do: help those creatures who could not help themselves. She had just started cooking her dinner when she noticed a small card sticking out from under the front door; she must have dropped it when she brought the mail in. As she was waiting for her frozen entrée to finish cooking in the microwave, she grabbed the card and examined it. It was a business card for a tree removal service. She looked at it for a second and was about to toss it in the recycle bin when her fingers felt something on the card. She held the card up and tilted it under the ceiling lamp. She could see a faint embossed symbol across the middle of the card, but the glossy coating and graphics made it difficult to see clearly. She ran her fingers over the embossment, tracing the shape and realized it was something she recognized: the teardrop symbol with the partial circle in the center, signifying Quadrant 1 of Robostadt. It had been more than a year since her adventure in Robostadt had ended, and while she was reluctant to go back so soon, she felt that the robots wouldn't be doing this unless there was a genuine problem. Her dilemma wasn't a matter of whether she would help, but how quickly she could leave her current commitments behind. She decided to eat dinner, relax, and sleep on it. The next morning, she called her mother, who lived a few blocks down the road, to inform her that she needed to go visit some friends who needed help, and asked her to watch her place and feed her animals while she was gone. Her mother agreed after double checking that she still had a copy of the key to Ash's house.

Ash grabbed a rolling suitcase she had previously filled with out-of-fashion clothing that she no longer wore, along with her uncle's dusty old map and was just about to jump in her truck to drive out to where the lift was located, when she realized she hadn't actually marked the location of the lift she used to leave the robot city. She only had the location of the broken lift and handwritten instructions on how to reach her truck from where she had surfaced. There was a second teardrop symbol on the map, but she had learned that it was an entrance for Quadrant 3 and she wasn't about to mess with that. She spent the next 20 minutes using the scribbled instructions to backtrack to where the lift was supposedly at. She decided to lightly mark the spot in pencil just in case she got it wrong. Ash said goodbye to her animals, petting and kissing each one, then she jumped in her truck and headed out.

Ash drove out to the old wooden shack she recognized and pulled her truck out of sight behind the structure. As she was getting out of her truck, she glanced up at the chimney and smirked. An old rotted piece of wood had fallen and had exposed the concealed upward-facing fisheye camera the robots used to track the sky colors. She didn't have a ladder or tools to fix it immediately, so she figured she will just have to tell the robots about it once she got to the city. She grabbed her suitcase and walked around the shack to the front door and knocked, just in case it wasn't empty. There was no answer, so she opened the door and walked in. With her flashlight, she made a cursory sweep around the room and then shined it over onto the fireplace. She immediately became irritated that someone at some point had brought in some logs and lit a fire. She grabbed the old straw broom in the corner and started sweeping the charred pieces of wood and carbon dust out of the fireplace. She swept the debris up into an old black iron dustpan and dumped it in the rusty ash bucket sitting nearby. Grabbing the fire poker hanging on the wall, Ash used it to pry up the large rusty plate in the bottom of the fireplace, then drug the plate out onto the floor, exposing a square concrete lined shaft with a permanent steel ladder. She backed her luggage over into the fireplace and lowered it down by a rope she tied to the handle until the rope went slack. She then climbed into the fireplace and while balancing herself on the ladder, she grabbed a peg “handle” welded onto the back of the plate as she continued to descend, sliding it until the front edge of the half-inch plate snugly fell into place. She continued down the ladder into a concrete room similar to what she saw previously at the other lift site, but this time, the lift was much smaller but in good condition and even had an inspection tag hanging on the wall with a recent date stamped on it. She smiled and flashed the tree service business card at the security camera in the far corner as she slid the gate closed and confidently pulled the lever to start the lift.

Ash stood waiting for the lift to finish its journey, slowly traveling the equivalent of more than 30 stories down into the darkness. Once it arrived at the bottom, she pulled the gate open and stepped out, shining her flashlight around. In a flash, she realized that a black sentry was standing next to the lift entrance less than a meter from her. She shrieked loudly and reflexively jumped back into the lift, dropping her flashlight. She frantically tried to pull the gate closed when the sentry stepped up and said “hello” in a calm, deep voice. All at once, she was startled and confused. “spiders don't have voices, they don't have personalities, they only exist to haul robots off to robot jail”, she thought to herself. Suddenly noticing the tiny red pinpoint of light at the base of the sentry's neck, she froze – the sign of the mythical 'black wraith' that Sol had told her about – the terrifying sentry that attacked anyone that reacted to it in any way. She really thought Sol had been joking. She finally let out a hoarse “hello” and the sentry introduced itself as the Central Mainframe for Stability. She stood there trying to remember... when they were last in the Mainframe Collective, Mel had recited the names of the eight central mainframe computers, each governing an aspect of robot society: energy, network, the city, work, and so on, but ‘stability’ was not one of the names that she remembered hearing. She asked the mainframe about its name and mentioned the wraith myth, and it responded with laughter. It indicated that it wasn't one of the eight, but a ninth mainframe sitting in the center of the eight. Its job was to pull together and examine data from the other mainframes as well as other sources and evaluate and maintain the stability of robot society. It also said that while the black “wraith” sentry did actually briefly exist, it was an attempt at upgrading a regular black sentry with one of the first generation-2 CPU modules. When activated, the sentry went berserk and fried two-dozen robot staff with EMPs and killed a human before they had lured it into one of the deep concrete-lined waste storage pits and waited until its battery power had been depleted. Once they were sure, they pulled its CPU module for analysis and put the chassis in storage. After reading a report about the incident and the unused chassis, the mainframe decided to use it as a remotely controlled “avatar” to keep a closer eye on Robostadt.

The mainframe then turned to talking about the mission. “Robostadt has a serious and persistent technology leak, with chips including actual CPU modules ending up in human technology researchers hands. As far as we can tell, all of the stolen CPU chips so far have self-destructed, as they are designed to do upon getting too far away from the city. We have had several instances where one of the human governments came close to discovering the robot city, which would certainly result in the fall of robot society and the destruction or subjugation of the robot population. We need to find and stop this leak, and bring whoever is doing it to justice.” Ash asked if her cargo bot friends could help, the mainframe stated with certainty that nobody involved with research can know about this mission, since any of them could be involved in the leak, even unknowingly. In realizing how critical this was, Ash asked how she would be able to help.

The mainframe responded, “Just as I remotely control this sentry, I am giving you a remote console with four remote “golem” modules. You can plug these modules into any chassis with an empty CPU slot in Robostadt and use grid programming to control the chassis. These remote modules will not have real robot personalities, however they will be able to do rudimentary things like install or remove a remote module in another chassis or follow some basic travel instructions along a public route.”

(story continued on next panel)

(“part 2” story, continued)

The mainframe continued, “You might need to find a way to explore the other quadrants, but be warned, you can send your remote controlled robots into quadrants 2 and 4 but you won't be able to go there yourself due to the extreme temperatures at those depths. The robots will be able to autonomously follow public pathways and use public transportation to destination points, but they will not be able to go 'off the beaten path' as humans say, without your grid programming.” Your remote console will have a map of the quadrant pathways to aid in directing the robots to the proper destinations, and a microphone to perform disguised vocal communication if you need to talk to other robots to get information. The remote console can also act as a terminal, so you can plug it into any public communication terminal to check the email account hardcoded into the console; If I get any more information, I will send it via email, so be sure to check the account regularly. Did you bring your harmonics band?” Ash briefly raised her arm exposing the band on her wrist.

“I have coded your band with an updated express pass to travel throughout Quadrant 1 and it is also coded with a key to human apartment 301 in the building west of the old lift down in Lower Robostadt. I have left the remote console and the four remote modules in that apartment. For now, that will be your permanent apartment in the city. The harmonics band will allow you easy access to move around Robostadt, however I recommend that you disable it while you are investigating leads, to reduce the chances of someone tracking you.” Ash smiled as she rolled her luggage out of the lift. “Just one more thing,” the mainframe quietly said, “Your uncle was very respected here – he did many a great deed for Robostadt and helped us survive while this city was being established. He was one of the three human principals of this city. He had hoped to bring you in to assist after his diagnosis, but you were away at schooling and unfortunately his plan didn't work out. We hope you can help us as much as he did. Good luck, and let me know via email if you need anything.”

The mainframe said goodbye to Ash, its sentry avatar disappearing into the shadows.

What needs to be determined:

1. Would this game expansion be fully linear? Open world? Linear major locations with multiple ways to get through each location? (i.e. Deus Ex)
2. Robot programming puzzles are critical and central to this game. How would the be incorporated into this expansion?
3. It may not be possible to do this expansion without a major rehashing or reuse of the same puzzles from the first story. To what degree would that be acceptable?

Stage – Not Implemented

Possible “part 2” story timeline:

1. Ash receives a business card indicating that she is needed in the robot city.
2. She finds her way to the lift and descends into the city.
3. The Central Mainframe of Stability meets her at the bottom of the lift using a black sentry chassis as an avatar.
4. The mainframe explains the chip theft situation and indicates that Ash’s harmonics band has been updated. Ash agrees to help investigate.
5. Ash heads down to Lower Robostadt and enters the locked apartment using the band. She retrieves the items the mainframe left for her.
6. She checks her watch and sees that it’s already dinner time. She briefly heads out to one of the garden spots and retrieves fruits and vegetables for a salad. The apartment is also stocked with canned goods and some frozen meals in a chest freezer in the corner of the living room, something the last Lower Robostadt apartment she stayed in didn’t have.
7. She examines the console and sees that it appears to have been altered to have a 3-D printed back cover with cavities for the four 'golem' modules to snap into, allowing for easier carrying. After unpacking her luggage into the bedroom closet, she connects the console to a public terminal port in the apartment and reads up on what has happened in the city since she was last there, eventually falling asleep on the couch.
8. The next morning, after eating and showering, and changing into a "nice" 60's outfit her mother gave her, Ash checked her email. She received a message indicating that she should check the library in Lower Robostadt. A robot reported witnessing a human giving the librarian what looked like a one-ounce gold coin for something small in return. She wanted to go to there anyway to refresh her programming skills. She grabbed her console and headed out the door, locking the apartment. She is surprised by the bustle of robots heading to work – something she never saw the last time she was there.
9. She enters the library, and notices that the attendant has not yet arrived. She steps up to one of the terminals and starts the programming tutorials. After she finished several sections, the attendant arrived and said hello, and she immediately cornered them, telling them that she knew about the gold coin. The attendant quickly looked around. It indicated that a human needed some complex business analysis software written, and had agreed to use the gold coin as payment. The attendant said that while this was not strictly illegal in Robostadt, it was taboo and highly embarrassing for a robot to engage in such human activities. Ash agrees to keep quiet about it, but asks the attendant to scan back through their offline memory storage for any humans acting suspiciously in the past year and email her if they find anything. The attendant agrees. Ash goes back to finishing the tutorial she was working on and then leaves. She starts walking down the street, away from the human apartments and human services building.
10. The first time Ash was in Lower Robostadt, right after she first arrived in Robostadt, Viz told her that Lower Robostadt streets formed a giant circle with buildings on both sides of the street, and alleyways connected Lower Robostadt to the industrial district. In the center of the circle, offset by about 2 meters upward was the lowest level of the industrial district. The ring Lower Robostadt was constructed in was originally temporary storage for groundwater, directing water to deep diagonal channels used to drain the water away from the city. Once the ring was drained down to an ankle-high level, the apartment building construction began. The stooped building design was chosen just in case seasonal excess groundwater became a flooding issue again. Currently, all excess groundwater is first sent up to the power plant to use for cooling, then it is sent down into Quadrant 2, where it is purified and distributed to the rest of Robostadt. The cluster of buildings closest to the old lift are structures containing facilities designed for humans: apartments, library, storehouse, medical clinic, gym, workshops, restrooms, etc. Other residential districts have similar areas.
11. Ash continued walking for what seemed like another mile, until she finally completed the loop and came across where the old lift she originally arrived in was located. It looked like the lift had been deconstructed, but the shaft was still open, and a large air grating was embedded in the ground behind the lift shaft. As she was examining the area, a bipedal robot that had recognized her approached her and said “Hello, I’m Relafa”. It wanted to thank her for certifying it in the Bot Proving Grounds last year. It indicated that it has kept her grid programs intact as a memento, since no previous robots had ever been certified that way. Ash asked it how it knew about that, since it had not been programmed with a personality at that point. It indicated that even without its personalization, its CPU module was still recording memories. Relafa said that the first thing it did on its own after personalization programming was to go back through and replay those memories. She asked it where it worked now, and it indicated that it was a security guard in the industrial sector. It apologized and said that it didn’t have much time at the moment, since it was headed to work, but it spelled out its name to Ash, just in case she wanted to keep in touch. “It was nice to meet you”, Ash responded, and shook Relafa’s hand and said goodbye. She then walked back to her apartment to have lunch.

City origin background (for possible story arc):

1. In 1906 a spaceship that had been traversing the galaxy looking for a suitable planet appeared in the solar system and discovered the Earth. Upon seeing that it was populated, it scanned the planet for its technological development and natural resources. After determining it was a suitable planet, it found a compatible solid granite mountain in a remote area of northern Maine and sent a crew to carve out a cavity for the ship. The ship then lowered itself into the cavern. A large keystone was placed above the opening, sealing the ship inside the cavern. Additional large shaped stones were placed at the sides of the ship, sealing the lower engine into its own cavity.
2. Its crew of 32 robots were a military contingent which had fled an underground robotic city that had been defeated by the native surface population. The crew's plan was to locate a planet that was at a suitable technology growth potential and quietly help steer the planet's technology development, while secretly building a robotic society that they could slowly introduce to the native population of the planet. They would build robots similar in appearance as the native population and even try to incorporate some of the architecture, customs, and languages into its own society in order to make their eventual reveal more palatable for the native population. The plan would be implemented very slowly and carefully so as not to risk premature discovery. They don't want to make the same mistakes their predecessors made that led to the fall of their previous robot civilization. Their plan was estimated to take between 120 and 150 years to implement. The general robot population of the city knows that it isn't logically feasible to assume that their technology came from the humans, but they have not been told exactly where their technology originated. This has led some robots into believing in some type of alien origination, time travel, or even parallel universe theories.
3. The black sentries of Robostadt are the crew of the spacecraft, which is why they look so alien. They act the way they do to conceal their plan and their capabilities and keep an eye on the city. The "black wraith" sentry incident was an attempt by the Robostadt researchers to hijack, drain the batteries, and forcibly upgrade one of the black sentries to use a newly-created CPU module. This ended up triggering a brutal military defensive mechanism built into the chassis itself when they went to recharge the chassis. This led to the loss of 20 research robots and a human being killed. Unlike all of the other robots in Robostadt, the black sentries have heavily shielded CPU modules that cannot effectively be scanned by Robostadt technology, which has always angered the Robostadt researchers. The "black wraith" sentry that was ambushed had a red pinpoint light at the base of its neck signifying that it was the captain of the spacecraft – after the incident, it was covertly rescued and restarted by the other black sentries. The "black wraith" sentry is not an avatar of a central mainframe, as it claims – it operates in its own capacity and monitors the eight central mainframes. The grey sentries are duplicates that the researchers have created, but are just normal robot chassis.
4. Lex's upgrade: During the "black wraith" incident, some flakes of the sentry's coating were obtained and quietly but painstakingly reverse-engineered in quadrant 3. Quadrant 3 researchers can now create a similar coating but need to rely on humans to purchase and transport some rare and very difficult-to-obtain materials.
5. The "nuclear reactor" in the center of the city is actually the forward part of the intact spacecraft. The plant workers supply the "reactor" with mineral oil to transfer heat into a heat exchanger, which in turn runs external turbines via a water/steam loop. The workers have been told that the reactor is fully automated with advanced robotics and doesn't need maintenance. In actuality, the ship's crew had tapped into the ships engine cooling loop and was using heat from the idling ship engine to produce electricity, since they did not have the technology to generate large amounts of electricity directly from the ship's singularity-based engine power source.
6. The black sentries and central mainframes are unaware that Lex has the shielded coating or that Quadrant 3 researchers have reverse engineered the coating – they think Lex is just another robot with a cosmetic paint job.
7. The previous robot civilization fell due to many missteps made by the robot leadership. They chose a planet where the native population had not yet started their industrial age, the robot civilization grew too fast and they waited too long to make their presence known. By the time the native population was beginning to do large-scale mining, they discovered that there was no underground ore left anywhere on the planet's central continent. The robots had secretly mined out all of the reachable ore and replaced it with slag. The robot city had grown to hundreds of square miles and hundreds of millions of robots. Most of the robots were built, only to be powered off and left in giant storage facilities. The robot civilization fell when a contingent of the robot opposition joined forces with the native population and triggered a meltdown of the giant main reactor, the explosion destroyed the power facilities of the robot city, starving the robots of electricity and the nuclear fallout blanketed the planet, resulting in a global extinction event for the native population. Several of the city's military ships were able to flee the planet before all the electrical power was gone.

Other story ideas:

The rescue (after story part 2 intro): After the explosion had occurred, Ash attempted to rescue the black sentry which was hanging above a deep fissure. Ash secretly knew about the black sentries being the crew of the spacecraft, but hadn't told anyone else what she had discovered. Ash grabbed a long chain and wrapped it around one of the support columns and tied a crude knot, then she threw the other end down into the fissure. After getting frustrated that the black sentry was "playing dumb" and ignoring her attempts to have it grab the chain, she blurted out a vague phrase about the black sentry's importance to its crewmates and their plan. The black sentry suddenly locked its crimson gaze on Ash, and she squirmed uncomfortably, not knowing if that was a gaze of impending death or a gaze of attention and acknowledgement. She again tossed the chain to the sentry and with a lightning-fast motion, it grabbed the chain, clamping it tightly with two of its grippers. She yells at some other robots to help her pull on the chain. As the robots begin to pull, thinking they were rescuing a regular robot, she whispers to the sentry that she supports its plan and is on their side – they can trust her to keep their secret. The black sentry's red gaze remained firmly focused on Ash, but its eyes had a strange flicker of blue lines, like some type of scan... or... communication was occurring. Finally, after another minute of pulling, the black sentry was back on solid ground, and the other robots, having realized what they had just rescued, dropped the chain and fled the area. Ash was now standing alone in front of the black sentry, face to face, with it towering a meter over her head. She proceeded to have a very awkward conversation with it.

* Robot personhood:
In Robostadt, a "robot" is a CPU module that has been given a personality profile and loaded with the Personalized Robot Intelligence Matrix software. If a CPU module is not personalized, it is just considered a machine/computer/device and it is not considered a robot. Robots usually always have their own simple bipedal chassis for off-work or full-time use, and some have one or more specialized chassis for work. Most robots leave their work chassis at work and swap into their personal chassis at the end of their work shifts. CPU modules have a small integrated battery and wireless chassis connectivity that allows a robot to pull its own CPU module from one chassis and install it in another without requiring assistance. Most personal chassis are given random cosmetic color combinations at the factory to make identification easier.

All personalized CPU modules in Robostadt have unique names. Their name is an index for their personality profile, which is also unique. Generation 3 robots use random Latin script names, with shorter names indicating earlier models. Gen 1 and gen 2 robots use alphanumeric codes, such as "is130" or "hm559". Since generation 4 robots are still classified as experimental, they use a combination of a greek letter and a 3-digit numeric code, such as "Epsilon 416" or "Psi 207", which is tied to a file in Quadrant 1 Research. For all robot generations, a personality profile is created by taking a standard personality optimized for a particular job function, and adding a slight "random noise" to the parameters. The personality is still ideal for the particular job, but the randomization helps prevent CPU logic lockstep conditions from occurring between robots working on the same job, and increase the likelihood that flaws or mistakes or sources of risk are discovered and corrected.

* Robot identification:
Most Robostadt robots have an acceptable machine equivalent to human vision, but one which can be easily confused by darkness, flashing lights or unusual light coloration. To overcome this limitation, they have various methods including spacial sensory input that detects thermal, electromagnetic, and piezoelectric, radar or laser reflections, stereo depth estimation, and other attributes to uniquely identify other robots, as well as identify interfaces, doors, walls, and humans. To make robot identification easier, each CPU module produced by Robolithography and Nanoscale Assembly is customized to have a set of low-level harmonics that are always emitted while the CPU is powered on. These harmonics are unique for each CPU module and can conclusively identify the robot or machine. Black robot sentries do not naturally emit these harmonics (apparently due to their electromagnetic shield coatings) and while they are identifiable as sentries (because of their non-electromagnetic properties), the sentries cannot generally be differentiated from one another. Lex also has this same strange attribute, but uses its radio antenna to broadcast its CPU module harmonics when necessary. In darkness, Lex can disable its harmonics and slip past most older robots without being noticed.

* Robot power:
The robots of Robostadt usually recharge during nighttime hours when factories are not in operation and plenty of power is available. Gen 2 and newer robots are able to utilize energy cells to partially bypass their regular power source and use the energy cell for their peripherals. They are able to adjust the energy cell voltage output to as low as 50% (for slower, more gentle and precise gripper capabilities) to as high as 150% (for faster movement and stronger grippers). Most robots have an energy cell slot in their configuration panel. After an energy cell is depleted, robots can drop them into recycling bins distributed around the city. Some gen 1 robots have an upgrade installed allowing them to use energy cells. Generation 4 robots (and other robots with a peripheral regulator upgrade) can adjust their peripheral power strength without the need for energy cells, but might still use energy cells for emergency power capacity.

* Robot programmability:
All modern Robostadt robots have their core robot programming contained within their CPU modules. Most robots have a secondary programmable area that is a holdover from when they were undergoing initial unit testing. Some employers utilize the programmable area to provide specialized programs used to aid a robot with repetitive tasks or in hazardous or sensitive environments. Some of the newer gen-3 robots feature a hexagonal programming grid rather than the usual square one, and some exotic machines (like gen-4) feature a triangular programming grid and are the most challenging to use.

* Robot configurability:
Your robot friends have a configuration panel underneath their programming screen. This configuration panel shows various configurable attributes of the particular chassis in use. A status page is also available on the console to indicate what opcodes and flags are available on the specific robot chassis along with a description to aid in programming the robots.

* Robot generations:
The first generation of robots are now considered obsolete, although most still have simple jobs to perform, such as hazardous environment, ore mining and processing, material recycling, and simple fabrication. These robots have been whittled down to smaller numbers due to the dangers of their work environments. The gen 1 robots primarily occupy Lower Robostadt, with a few residing in midtown or work as assistants for gen 3 robots. There is some arguments about whether the black sentry "spiders" are really generation 1, or if they are part of a generation of robots preceding what are now called generation 1 robots. Approximate generation 1 population in Robostadt: less than 1000 robots. Black sentry robots: 32.

The second generation of robots work in various middle-class commercial jobs. Some upgrades have been developed for gen 2 robots, allowing them to be in jobs related to hazardous environments or requiring increased gripper agility, strength, or faster movement. Gen 2 robots mainly occupy midtown, although a few live in Upper Robostadt or Lower Robostadt. Approximate generation 2 population in Robostadt: 1500 robots.

The third generation of robots have been developed for more advanced roles requiring increased physical capabilities such as speed, strength, and agility, as well as higher levels of reasoning and creativity. Their faster CPUs give them more cognitive abilities and utilize more advanced AI engines. The Gen 3 robots work in advanced roles in research and high-tech commercial environments. Major upgrades are uncommon for gen 3 robots due to how new they are, although smaller sensory, cosmetic or peripheral upgrades are readily available. The gen 3 robot neighborhoods are spread between Midtown and Upper Robostadt, with a few choosing to be in Lower Robostadt. The Robostadt government is currently gen 3 bots. Generation 3 population in Robostadt: 2000+ robots.

The forth generation of robots, currently in experimental status, is the newest generation, with very few robots currently in full operation. These robots feature the most advanced AGI engines with the highest-performance mega-CPU modules and nearly unlimited memory. Their official jobs are in consulting for research, commercial, and government sectors. Unofficially, they excel in creating fiction e-books, puzzles as well as simulation performances and video games. Most have secondary work in entertainment directed towards the gen 2 and 3 robots. The quadrant 1 gen 4 robots have temporarily been placed in the Skyline district to isolate and better monitor their activities, since all so far have been affected by hyper-narcissistic processing disorder. Gen-4 robots feature a triangular programming grid interface that is purposely very difficult to master. Official generation 4 population in Robostadt: 20 robots in Q1 and one robot each in Q2, Q3, and Q4. Unofficially Q3 has 5 additional generation 4 robots; It is unknown as to whether these 5 were taken out of the documented Q1 population. As experimental units, the Quadrant 1 gen 4 robots are regularly wiped and re-personalized.

* Robot/CPU chassis classes:
Class 1: Bipedal, <3m height, <1m circular footprint, locomotion: mechanical legs
Class 2: Cargo, fits within 2x2x2m area, locomotion: plasma thrusters and hover pads or motorized wheels
Class 3: Fixed/Track, no size restrictions, locomotion: track, wheel, chain, or cable drive
Class 4: Fixed/Non-movable, no size restrictions, locomotion: none

Chassis tiers:
Tier 1: 25Mhz interface, 1K visual resolution + radar, suitable for any generation of robot.
Tier 2: 50Mhz interface, 2K visual resolution + radar & group sensory cross-transmissions, suitable for Gen 2+ robots.
Tier 3: 100Mhz interface, 4K visual resolution + radar/lidar & area/group sensory cross-transmissions, suitable for Gen 3+ robots.

* Slang:
"Android" is slang for personalized bipedal robots.
"Droid" or "Sub" is slang for non-personalized bipedal robots. Calling an android a "droid" is considered an insult.
"Super" slang for generation 4 or experimental robots.
"Max" is slang for a security guard, sector police officer, or gray sentry.
"Spider" is slang for a black sentry.
"Tracer/Tracker" is slang for robots which operate chassis that are fixed to tracks or follow hidden guides.
"Anchor/Anvil" is slang for any robots which plug into computers or machines for use as a CPU source.
"Slums/The Slums" is slang for the Lower Robostadt residential district.

* Value:
Each CPU module has an embedded 500g "gold coin" plate. While this plate doesn't add any electrical or thermal benefits, it does add minimal shielding benefits to the module, and is primarily used to add inherent value to the CPU module. The plate cannot be removed without destroying the CPU. This does increase the risk of humans stealing the modules to melt down for profit, however the mainframes felt that this "inherent value" was important to the robots.

* Humans in Robostadt:
Humans are uncommon in Robostadt, but they do occasionally pop up or are called upon for assistance. Most robots consider them a curiosity more than a threat, and security robots are programmed to specifically try to monitor and maintain a visible presence around untrusted humans during their visits to reduce illegal chip tech. sales between robots and humans. Most humans that know about Robostadt are archaeologists or other scientists that are willing to keep it secret. Some robots practice gardening as a hobby and are delighted to provide food to visiting humans. Toilets and wash rooms are also available on each residential level for human use. Some humans are trusted enough to have special roles within the Robostadt community. Because Robostadt doesn't currently have a native source for aluminum (bauxite) ore, humans play a very important role in covertly exchanging excess gold ingots for large aluminum ingots used to manufacture new robots.

* Breaking rules:
Robot jail: robots caught breaking robocode can be remotely deactivated or can be overridden to force them to walk to the nearest police location or courtroom. Robot security will be more lenient about humans breaking some rules, but may move the human out of a secure area or ban them from a city level. Humans found to be distributing illegal chips will be escorted to the surface exit and will be told to leave.

* Autonomous summons:
A robocourt judge or central mainframe has the option of issuing an autonomous summons for gen 3 robots to appear in person for more serious charges or when a robot has repeated ignored previous summons. The summons is sent out as a coded radio signal that is decryptable only by the specific robot recipient of the message. The successful decoding of the message will result in the CPU module's chassis output control functionality being disabled temporarily, and the underlying low-level chassis AI getting a command to take the robot to the specific location on the government level. Only then will the chassis functionality be restored.

Robot Console Screens:

Opcodes

Details

Program

Configure

Inventory

Exit

Opcode Listing

Opcode Table

Opcodes

Details

Program

Configure

Inventory

Exit

Opcode list with favorites

Zoomable/Scrollable Programming Grid

Scrolling Version History

Opcodes

Details

Program

Configure

Inventory

Exit

Chassis: Standard Compact Cargo Robot

Movement

THR,N,OFF/THR,N,ON THR,F,OFF/THR,F,ON

THR,E,OFF/THR,E,ON THR,R,OFF/THR,R,ON

THR,S,OFF/THR,S,ON THR,B,OFF/THR,B,ON

THR,W,OFF/THR,W,ON THR,L,OFF/THR,L,ON

THR,A,OFF

DMODE,C/DMODE,R

RRROT,R/RRROT,L

Other Controls

GRIP,OFF/GRIP,ON

Interfaces

RREG,RD/WREG,RD

IFC,OFF/IFC,ON/RREG,IFC/WREG,IFC

RREG,MC/WREG,MC/MCRST

Opcodes

Details

Program

Configure

Inventory

Exit

MC

Key

EM SHIELD

Backpack Inventory Screen

Opcodes

Details

Program

Configure

Inventory

Exit

MC

Key

EM SHIELD

Storage Area

Area

EM

Contact

Gripper

Directional

Auxillary Control

Opcodes

Details

Program

Configure

Inventory

Exit

Input Flags

32 NB b0 North/Front Bumper

33 EB b1 East/Right Bumper

34 SB b2 South/Back Bumper

35 WB b3 West/Left Bumper

44 GA b12 Gripper Active

45 IA b13 Interface Active

46 AS b14 Area Sensor (electromagnetic)

47 CS b15 Contact Sensor (electromagnetic)

62 MC b30 Memory Card Sensor (MC1-,MC2RW)

Input/Output Flags

48 NT b16 North/Front Thruster

49 ET b17 East/Right Thruster

50 ST b18 South/Back Thruster

51 WT b19 West/Left Thruster

- All possible sensor module types:
- Area Only Module
 - Contact Only Module
 - Directional Only Module
 - Gripper Only Module
 - Area/Gripper Module
 - Area/Contact Module
 - Contact/Directional Module
 - Directional/Gripper Module
 - Contact/Gripper Module
 - Area/Directional Module
 - Contact/Gripper/Directional Module
 - Area/Contact/Directional Module
 - Area/Contact/Gripper Module
 - Area/Directional/Gripper Module
 - Universal Module

Robot sensor configuration panel:

Area ☐ ☐ Contact ☐
Gripper ☐ ☐ Directional ☐

Note: Not all sensor module types will be used in the game. Different module types will be used to manage puzzle difficulty and reduce the potential for cheaty setups.

Cargo Bots:

* Cargo robot description:

The four robots the player controls are compact cargo bots (1.25m, roughly cube shaped), used to move materials for the research department. Three of them are silver with trim colors in red, green, and blue, and are named Sol, Viz, and Mel, respectively. The fourth cargo bot used to be silver and purple, but is now black with dark purple trim, and is named Lex.

* Lex's upgrade:

Lex has a full-chassis anti-electromagnetic coating and EMP weapons installed. The coating is very rare and expensive and can only be obtained from the third quadrant. To conceal this operation from the government, a human had to transfer Lex's chassis without its CPU module up onto the surface and then down into quadrant 3's surface access point a dozen kilometers away. The chassis had to remain in Q3 for several weeks to have the ferro-ceramic coating applied. After the chassis was returned, the CPU module was reinstalled and Lex then had to undergo Bot Proving Grounds re-testing. One of the Q1 research robots divulged to Lex that they payment for his upgrade was five prototype generation 4 androids that were transferred to Q3 at the same time as Lex's chassis. This whole operation had to be carried out at night, without the government's knowledge and without being seen. A very expensive upgrade, indeed.

* Cargo bot misc dimensions:

Cargo robot size: 1.2m cube plus 40x68cm cylindrical head (excl. hat rim width) and 1.5m shoulder to shoulder width. When cargo robots are stowed in backpack or shut down, they retract their heads into their cube chassis area for added protection and space reduction. Cargo bots have an integrated shrink beam allowing them, when empty, to shrink themselves down to an 18cm cube for storage. When they shrink, their shrink beam reduces their mass by an equivalent amount.

Cargo robot compartment: 1m cube interior with detachable ceiling-mounted touch screen.

Cargo compartment door: 1m wide by 75cm tall.

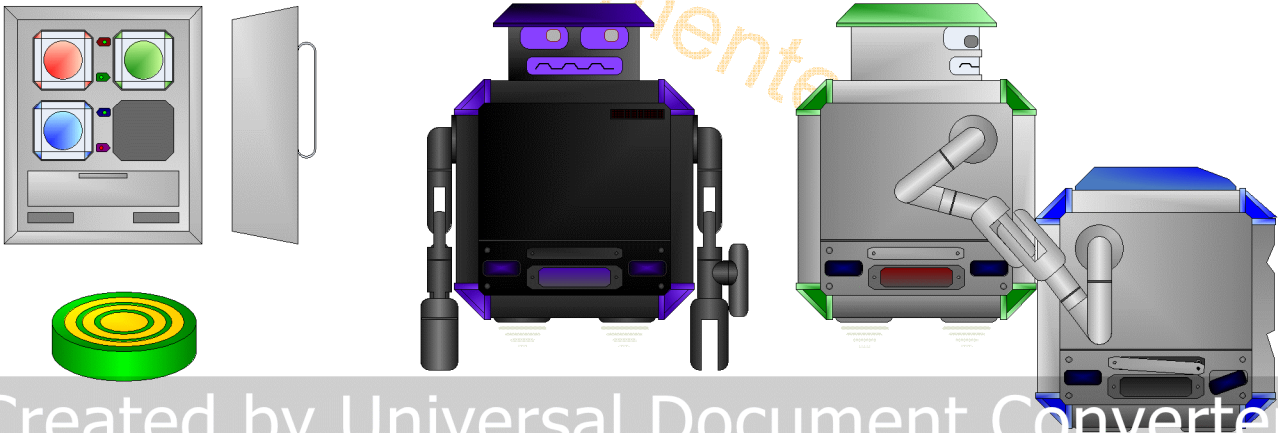
Carrier/backpack size: 20cm x 60cm x 75cm (approx. 8x24x29"). The carrier is aluminum, trapezoidal shaped and contains four slots for shrunk robots, two auxiliary energy cell slots for providing backup power to the robots, and a small tilt-out drawer for storing unused parts and cables. The carrier has straps on the largest surface for strapping into a cargo compartment or using the carrier as a backpack. It also has an integrated speaker to pick up the cargo bots private radio channel signal and convert it to verbal output. It has rudimentary machine vision so that shrunk robots can see what is around them before exiting the carrier. The carrier has a wireless console stored flush in a cavity on the back side just below the straps and a wireless remote control next to it. This backpack was made to fit bipedal robots, but should also fit humans.

* Cargo bot starting points:

You start by obtaining the backpack with the green robot in the Lower Robostadt. The red robot is hidden in the Industrial district. The black robot can be seen but is not yet accessible in the midtown district in one of the residential buildings. The blue robot is hiding in the commercial district. Once you get to Upper Robostadt, you get an access key which allows you to free the black robot in midtown. You should have all four robots when you enter the Robostadt Research district.

* Cargo bot trust:

For the four cargo bots, their trust in Ash will start at 50%. Trust can increase by selecting good NPC dialogue choices when talking to the robot, continuing progression in the game, good repairs that the robots are happy with, using a robot in a puzzle and having them press a critical button or grabbing a critical item, etc. Trust can decrease by a robot getting damaged, having an incident with a sentry, not being repaired promptly, receiving a poor repair job, running completely out of power, bad NPC dialogue choices, etc. Extreme high or low trust from one robot might also affect the trust level of other robots due to the closeness of their friendships with each other. Lex is especially sensitive to damage, not getting repaired promptly and low-quality repair choices due to its expensive and irreplaceable chassis. Mel is more sensitive to low trust levels of their cargo bot friends affecting their own mood, and being the one who solves the puzzle. Viz loves to interact, so it is more sensitive to good and bad dialogue. Sol is very mistrustful of authority, and will be dramatically negatively affected by interactions with sentries, especially black sentries, but might be positively affected by bureaucrats successfully resolving problems. The higher the trust level of a robot, the more likely they will offer NPC dialogue for optional puzzles, locations of hidden items, map shortcuts, useful hints during puzzles, and more robust dialogue for robot history and lore. If a robot's trust drops low enough, the robot may stop offering any optional dialogue at all, and may not be too friendly with their goodbyes at the end of the game. A robot's negative mood can also cause Hyper Interrupt Query Syndrome, which can result in excessive battery utilization even while idle, lower timing accuracy of programs, and decreased movement speed. A robot with this condition may not even be aware that they have it. The only fix for this condition is to improve their mood by increasing their trust level in Ash.



Game Concept: CPU² Robots (Continued)

Robot puzzle concepts and ideas:

Maze traversal using different algorithms
Inter-robot radio communication
Environment counting & sequences
Passing objects between robots
Recording and replaying sequences
Robot pushing or pulling objects to allow passage
Downloading programs which give you ability to solve puzzle
Robots accessing interfaces to obtain puzzle solution or hints
Utilizing a small programmable drone for hard-to-access or hazardous areas
Robots controlling a machine to manufacture a needed sensor or device
Noisy environments requiring radio communications with noise-resistant protocol
Hijacking/reprogramming other NPC robots or machines to overcome obstacles
Programming robots for mundane tasks to earn money for advanced programs, information, or sensory objects.

* Temporary robot upgrade possibilities:

- Upgrade robot to ¼-size drone chassis for confined space access. Two robots will be upgraded for this: one to a drone body, another will have its gripper swapped with a drone port. The gripper ON/OFF functions will be used to launch and land drone, and the drones programs start when it is launched and terminate when it lands.
- Upgrade gripper to laser to burn away weak walls, debris, and garbage.
- Add directional EM shield and EM sensor to robot to move robot through dangerous area with EM rays. If directional EM shield is activated, corresponding bumper and thruster will not be functional, so the shield might have to be selectively powered on and off to get through an area. EM sensor will indicate which direction EM rays are coming from, and puzzle might occasionally or frequently change EM ray direction.
- Upgrade replaces thrusters with wheels or tracks for areas where thrusters can't be used due to explosive gasses. Wheels or tracks might change dynamics of moving the robot around.

* Puzzle Obstacles:

Filter beam: Programmed to only allow certain entities to pass, such as only robots, only humans, or only sentries. Beam will be made up of individual colored elements which indicate restrictions.
Robot guards, turrets, directed EMP, locked doors, laser walls, electric arcs
Hazardous areas such as toxic, high-EMP, windy, seismic, icy, unstable
Moving objects such as walls, elevators and other robots
Invisible walls or paths requiring robot sensors to navigate
Pressure plates, wall switches, pushable blocks
Movable mirrors to redirect lasers
Access devices: Keys, Tokens, Tickets, and Passes
Robots can be drained of power with EMP, EM rays, or electric arcs. Battery cells can recharge a robot.
Robots can be damaged by EMP, strong EM rays, lasers, electric arcs, and turrets. Damage will cause thruster movement and gripper activation to progressively slow and expansion slots to become non-operational. Places will be available for robot repair, but will cost repair tokens.
Certain places will be hazardous to a human outside of a robot.
Certain places will be hazardous to a human even inside a robot.

* Puzzle rewards (for direct use or trading):

Wafers (flat currency, crafted from waste silicon wafers)
Energy cells and extended energy cells
Bronze, silver, gold, or platinum repair tokens
Robot upgrade authorizations
Zone transit passes
Surface Internet VPN hourly passes
Sensor modules needed for other puzzles or game progression
Programs or security codes needed for other puzzles or game progression

Sensor ideas:

Types: Area sensor: Report if an object is in the area (visible on the screen).
Contact sensor: Report if the robot is in contact of the object.
Directional sensor: Report the direction of the object in relation to the robot.
Gripper filter: Restrict the gripper to only picking up this type of object (object sensors only)
Specific object sensor: A sensor detecting a specific object.
"Any object" sensor: A sensor detecting any loose object.
Sentry sensor: A sensor detecting sentry robots.
"Any robot" sensor: A sensor for detecting any robot.
EM ray sensor: A sensor for detecting a strong electromagnetic ray presence or source.
Turbulence sensor: A sensor for detecting wind or other turbulence.
Compass sensor: A sensor for detecting the north pole direction (magnetic declination corrected)
Acceleration sensor: A sensor that will detect movement or movement direction.
Interface sensor: A sensor that will detect an interface location.
Track sensor: A sensor that will detect the presence or directions of a hidden guidance track

Repair token tier & repair time risk metrics:

	Repair token:	Platinum	Gold	Silver	Bronze
Slow repair (20 minutes):		N/A	0%*	5%	10%
Normal repair (10 minutes):		0%*	5%	10%	25%
Fast repair (2 minutes):		5%	10%	20%	50%

* includes 1 free follow-up fast repair for the same robot.
Percentage = chance or speed of degradation leading to possible future failure.

Filter beam/field colors:

Red: Robots not allowed
Blue: Humans not allowed
Purple: Humans only allowed if carried within a robot
Green: Health/capability check: Entities must move across the beam or through the field on their own.
No carried/pushed/pulled entities. Excessively damaged robots or injured humans are also not allowed to cross.
Green/Orange*: No assistance. The player must press the associated start button before the robot crosses the green beam and must keep button pressed until the robot crosses the orange beam, otherwise the orange beam will turn red and block the robot from exiting.
Black: Nothing is allowed to cross the black beam/fields, although the black beam/fields will be disabled if the robot successfully crosses the green/orange beam/field pair.
Any of the beams can be disabled with optional switches. Fields or beams may also indicate a one-way requirement with arrows or triangles.

Random robot certification attributes:

Movement:
standard movement: switchable north/south/east/west & forward/backward/right/left
compass directions only: north/south/east/west
relative directions only: forward/backward/sidestep right/sidestep left
unidirectional: forward/rotate right/rotate left
bidirectional: forward/back/rotate right/rotate left
crabwalk: sidestep right/sidestep left/rotate right/rotate left
broken crab: sidestep left/rotate right
reversed: same as relative directions, but right and left are reversed
interfaced: movement is accomplished via text inputs into the interface register.
Sensor:
full wall sensing: 4 wall sensors, compass or relative
front sensing only: front wall sensor only
inline sensing only: front/rear wall sensor only
relative sensing only: 4 relative wall sensors: front/rear/left/right
compass sensing only: compass wall sensors: north/south/east/west
crabsense: right/left wall sensors only
front distance sense only: 4 front distance sensors: 0m/1m/2m/3m+ (ds0,ds1,ds2,ds3)

Grid:

standard grid: square programming grid (80% chance)
upgraded grid: hexagonal programming grid (16% chance)
exotic grid: triangular programming grid (4% chance)

First 5 robots in line (always the same):

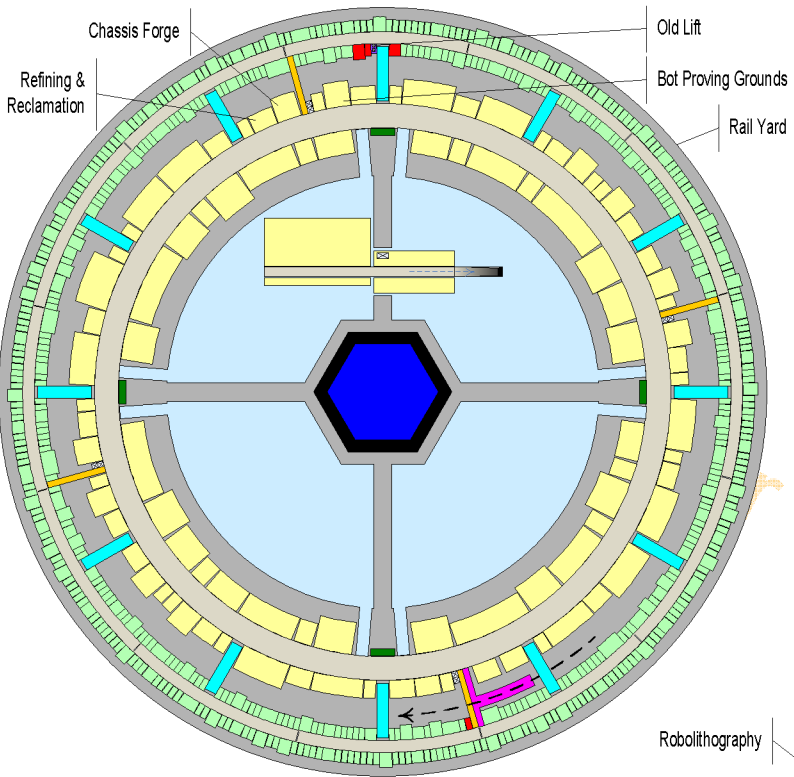
Standard cargo bot: standard movement/full wall sensing/standard grid
Upgraded cargo bot: standard movement/full wall sensing/hexagonal grid
Standard bipedal: relative directions/relative sensing only/standard grid
Warehouse picker: crabwalk/crabsense/standard grid
Wheeled delivery bot: bidirectional/inline sensing only/standard grid

Robot randomization uses a dedicated pseudorandom generator with seed chosen at the start of the game, so repeatedly reloading a saved game will not affect future random robot selection.

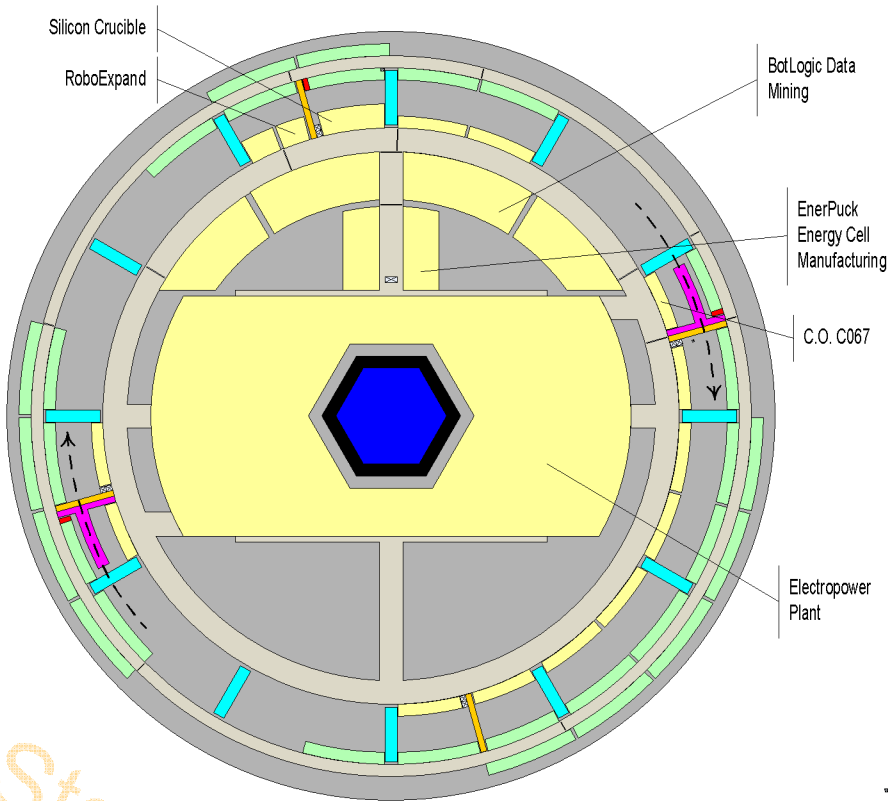
Robostadt City
Robostadt is the name of the robot city located underground in a remote area. The city is broken up into four quadrants, simply called Quadrants 1 thru 4, or Q1-Q4. This entire story will be located in Quadrant 1, with only vague mentions of the other quadrants in the story. The quadrants are connected via mining tunnels fitted with rails allowing the automated movement of raw and finished materials between quadrants, but does not allow for the movement of robots between quadrants, which is strictly forbidden.
The overall map will consist of eight to ten levels for the regular gameplay. These levels will alternate between levels with tutorials, story-related lore, and simple puzzles to learn some aspect of assembly programming and levels that have challenging puzzles that require programming features that the player has learned.
Quadrant 1 Levels:
• Lower Robostadt – cramped, a bit damp, filled with lower-class bipedal robot workers. Tutorials: thruster & gripper opcodes, conditional bumper flag functionality, simple loops, fork opcodes, radio. Street design style: ornate brownstone stoops • Robo industrial zone – Bot Proving Grounds, Chassis Forge, Refining & Reclamation, Base Logic, Lasenrich. Puzzles: simple mazes with varying wall configurations and grabbing objects, simple communication between robots. • Midtown – small robohouses filled with middle-class robots. Tutorials: basic register access, ALU functions, nested loops, interface I/O, memory card I/O. Street design style: large apartments with old-style façades. • Commercial zone – Silicon Crucible, BotLogic Data Mining, Robot Memory Analysis, ElectroPower plant, RoboExpand. Puzzles: timing puzzles, added sensors & functions, simple interface puzzles. • Upper Robostadt – large robot condos with lots of security. Tutorials: robot display I/O, memory & dupe opcodes, advanced ALU functions. Street design style: modern condominium buildings • Robostadt research -- Logic foundry, Robolithography, Nanoscale assembly, Automaton training. Puzzles: Interface puzzles, drone puzzles, black-box program puzzles, search and sort puzzles. • Skyline district – Penthouses with lots of exotic robots and machines. Tutorials: Macros, subroutines, page functions. Street design style: Individual glass-front houses. • Robogov district – District Management, City Control, Supercomputing, Mainframe Collective, City Monitoring. Puzzles: hybrid interface/maze puzzles and other advanced puzzles (regular end of game) • Drainage tunnels – filled with lost, corrupt, or criminal robots and organized crime rings. Tutorials: Miscellaneous remaining opcodes. Street design style: makeshift buildings built into cave-like areas. • The city's secret – restricted area below the city, with hazards and increasingly difficult challenges. Puzzles: Miscellaneous difficult puzzles.
Room configurations:
1. A single fixed room between 3x3 and the maximum static on-screen room size (such as 24x16) 2. Multiple identical-sized rooms with page flipping between rooms. 3. A long horizontally-scrollable room, 3-16 high and width more than the on-screen room width. 4. A long vertically-scrollable room, 3-24 wide and a height more than the on-screen room height. 5. A room larger than 24x16 in both dimensions, with both X any Y scrolling.

* Quadrant origination: Quadrant 1 is located in what was originally a limestone cavern sitting 300 meters below ground that stretched 350 meters vertically and 150 meters across. It was discovered when a low-grade copper ore mine south of the city had a tunnel dug at a 20-25% grade about 2 kilometers long that breached the side of the limestone cavern and ended up being abandoned due to immediate flooding and loss of life. The old lift down in lower Robostadt was discovered later on when surface surveys were carried out, and a 150m temporary tunnel was dug from just below the lift to breach the limestone cavern. This tunnel was used to direct and pump the water out of the bottom of the limestone cave, albeit causing significant damage to the old lifts metal support structure. Once the surrounding area was surveyed, a large coal formation was discovered sitting about a kilometer underground northwest of the city at a distance of 10km from quadrant 1. This formation would form the basis of quadrant 2. A tunnel was dug from quadrant 1 to 2 to facilitate discrete, unfettered access. While digging the access tunnel, A large gold bearing quartz formation was discovered and a branch tunnel was created in what would become quadrant 3, at about 500 meters deep below the top of a mountain. As test cores were being drilled in quadrant 1 to locate a source of bauxite, a large copper sulfide ore formation was discovered deep below quadrant 1 at about 2.5km. A small shaft was dug down into the formation and enlarged at the bottom to become quadrant 4. A large angled tunnel was dug from quadrant 4 over to the intersection of quadrant 2 and 3, and that intersection became a high-security rail transfer point for materials passing between the quadrants. The original shaft between quadrant 1 and 4 is still in use as a ventilation shaft near the original mining shaft connecting quadrant 1 to the surface. Quadrant 4 is currently creating an extreme deep shaft for further ore exploration.
* Quadrant manufacturing capabilities: Q1: Metal ore processing: silver, lead, nickel, zinc, and copper metals and alloys, uranium enrichment. Q2: Coal ore deposit: coal, coal coke, pure carbon, rare earth ores, and purified water. Q3: Large quartz/gold deposit: gold ore refining, raw silicon and quartz products. Aluminum** processing. Q4: Copper ore deposit: Ores: gold, iron, copper, silver, zinc, uranium, other trace elements. ** Robostadt does not have a native bauxite mine, so excess gold is covertly traded with a nearby human-run gold mine. The gold mine takes a cut for themselves and uses the rest to purchase large aluminum ingots for the robot city. Quadrant 3 has a hidden surface tunnel disguised as an old mining adit for transferring gold and aluminum via truck and for limited human access to the robot city. Each quadrant also carries out its own aluminum recycling.
* Quadrant advanced commercial & research capabilities: Q1: Silicon boules & chip production, advanced robot programming, robot chassis parts, nuclear power. Q2: Plastics, gasses, carbon composites, rare earth alloys, chemicals, emergency coal power. Q3: Fiber optics, industrial diamonds, glass lenses, sapphire glass products and ceramic coatings. Q4: Iron ore refining, steel production, metal plating, geothermal power.
* Human access Humans can access Robostadt via two shafts into Quadrant 1 and a surface entrance into Quadrant 3. Most humans can be found in Quadrant 3, with fewer found in Quadrant 1. Quadrant 2 and Quadrant 4 are not suitable for humans without protective gear due to extreme temperatures (up to 80°C/176°F). Robostadt is assisting humans in developing suitable suit technology for extended service in high-temperature depths. Heat exchangers are used to transfer heat from Quadrant 4 up to Quadrants 1 and 3 during colder seasons to balance temperature swings.
* City day/night cycle. At some point, humans helped the robots set up a surface camera on an old mining opening near quadrant 1 to obtain a fish-eye skylight picture with audio. Parts of the city use that information to control ambient lighting lamps to imitate the skies color and brightness, with some areas even being able to simulate sunrise and sunset colors and dark stormy days with actual rain noise (but not actual rain). Most robot workers follow a regular day work shift, evening time off, and nighttime downtime to recharge and run optimization and exercise routines on their robotics and to further their AI subsystem training.
Robostadt's central mainframes (the eight pillars of robotic society): Energy & Resources Networking & Computing City & Infrastructure Work & Production + 9th mainframe: Stability & Control Intelligence & Knowledge Accuracy & Quality Robotic Health & Growth Human Relations & Protection

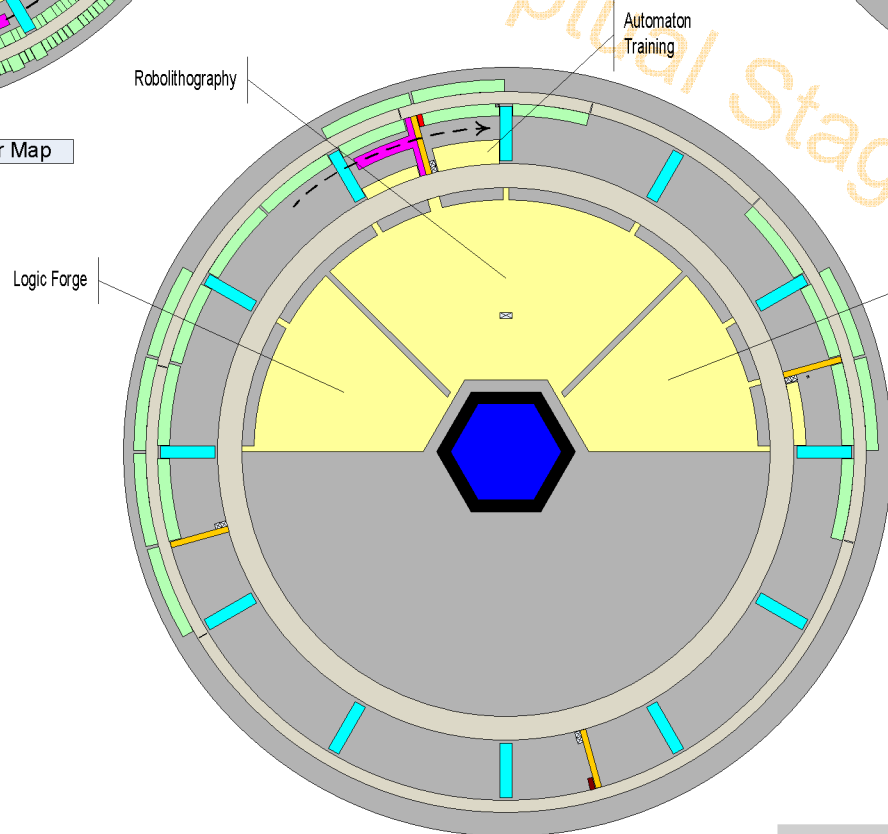
* Building addressing. Most of the buildings of Robostadt are located on single or concentric circular pathways. Addresses start with a quadrant specifier, Q1-Q4, or alternately the appropriate teardrop symbol. The levels are labeled by a single letter: A=Government sector B=Skyline district C=Research sector D=Upper Robostadt district E=Commercial sector F=Midtown district G=Industrial sector H=Lower Robostadt district (1 ring) Individual buildingsand structures are addressed incrementally, with 0 being the northmost building or structure, incremented by one. Decimals can be added as needed for structure locations being split into smaller structures. Concentric circles are then specified starting with 'A' as the innermost side. A concentric circle will have two letters, one for each side of the street. Floors of a building are labeled in hundreds starting at 100 for the first floor. Suites are labeled as the other two digits added to the floor number, or zero if the structure doesn't have rooms. The apartment that Ash first uses is "Q1H002A101", meaning "Quadrant 1 Lower Robostadt, Circle A, Structure 2, Room 101." The old lift is at "Q1H001", meaning "Quadrant 1 Lower Robostadt, Structure 1." Prominent structures such as accessways might not have the circle or room number designations.
* Travel restrictions Due to the nuclear reactor shutdown, large industry is mostly shut down in Robostadt, and robots are mostly confined to their residential areas. If robots are not working and do not have schedule-sensitive hobbies, they are urged to shut down until the restrictions are lifted or stay in sleep mode for the duration of what would be their work hours, in order to minimize Quadrant 2's emergency coal power plant utilization. The residential districts have each been segmented in 12 zones, with robots being disallowed to cross zones unless their work is in a different zone than their residence. The industrial, commercial, and research sectors each have all of their fire protection doors sealed, which would segment them up into dozens of sections of a few building entrances each. Some buildings have multiple entrances split between sealed sections, allowing limited travel between segments.
* Express Pass Access Four express pass accessways exist on each level, in the industrial, commercial, research, and government sectors. These accessways are at the 20, 110, 200, and 290 degree points on each level. Each Express area has two elevators. A triple-chained-car elevator that travels back and fourth between adjacent floors, allowing access between the industrial and commercial, commercial and research, and research and government levels. A second elevator with dual-chained-cars travels back and fourth across two levels, providing access between the industrial and research, and commercial and government floors. Using both of these elevators, a robot or human can travel from the industrial level to the government level, or vice versa. In addition to these modes of transporation, one subway loop is in operation. The loop travels clockwise from just after midnight until noon and counterclockwise from noon until midnight, and shuts down for maintenance at midnight, usually for about an hour. Subway stops: commercial at 75°, industrial at 165°, commercial at 255°, research at 345°.
Robots using express passes will have them inserted into a slot in their chassis and read electronically as they enter an express-travel area. Humans will have to carry around a console or terminal device with the equivalent slot. Robots using an express-travel service without a valid pass will receive a fine electronically. Humans will be stopped and asked for identification, then will be reported to the government. Humans with a harmonics band may have a permanent express pass coded into their band, either for limited access or full access.



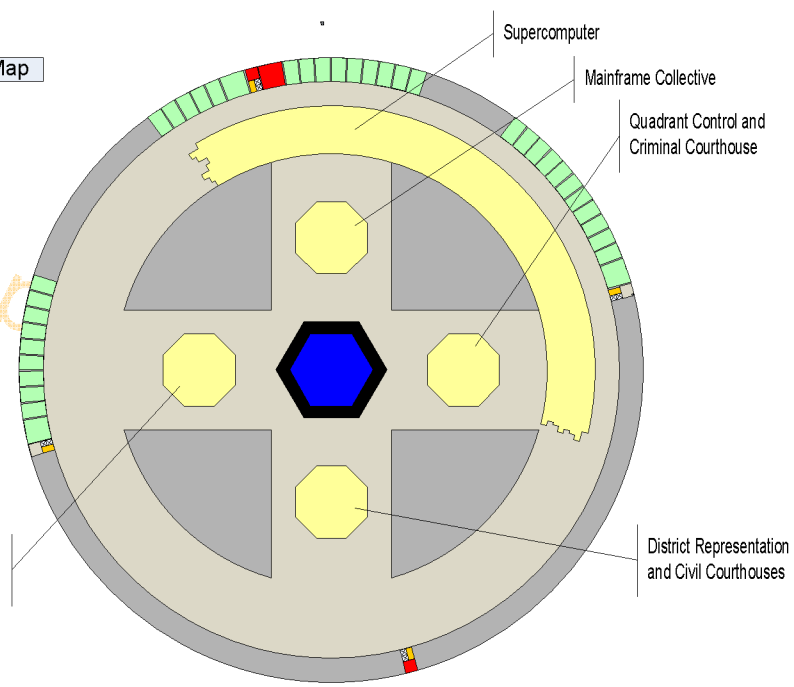
Lower Robostadt & Industrial Sector Map



Midtown District & Commercial Sector Map



Upper Robostadt & Research Sector Map



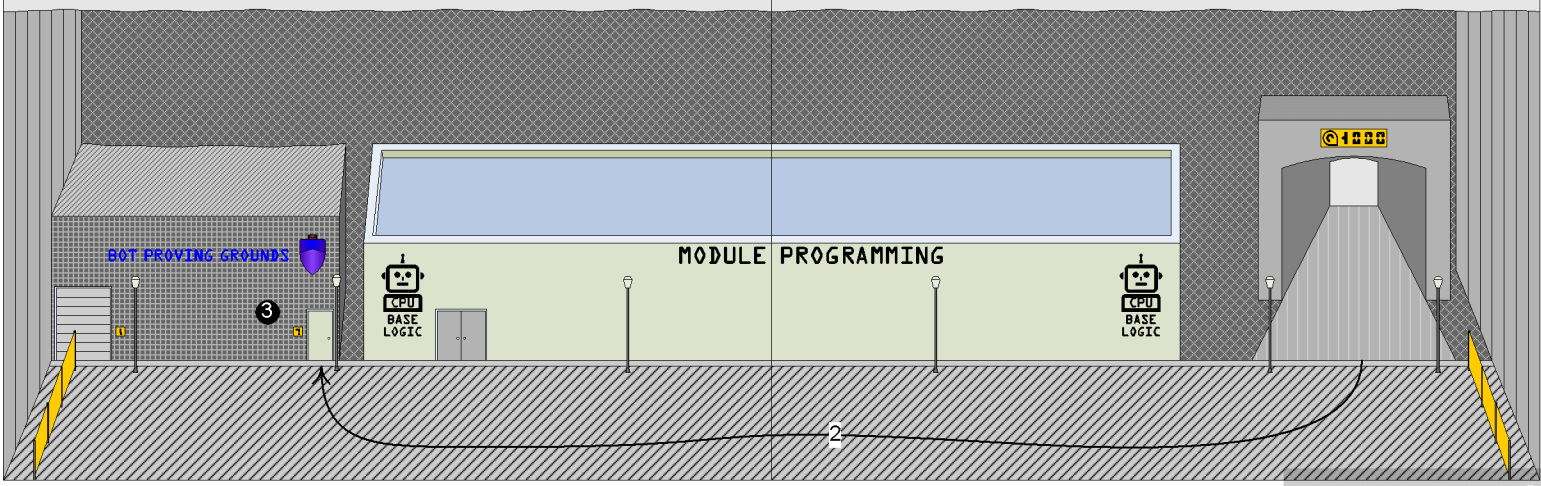
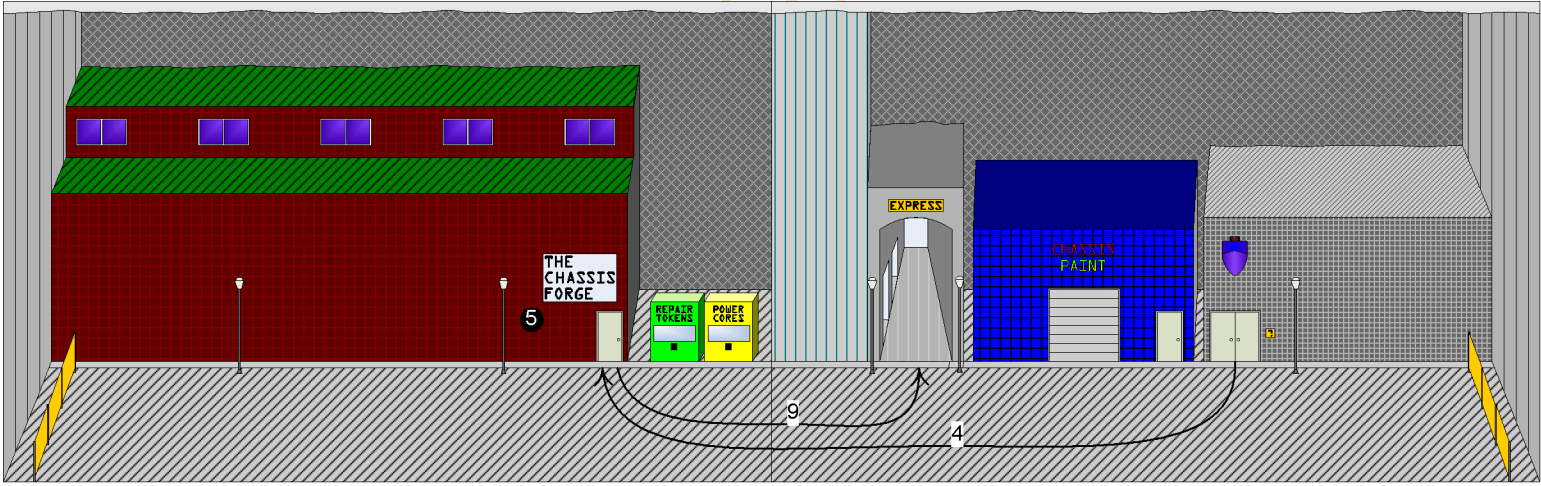
Skyline District & Government Sector Map

- Lower Robostadt District & Industrial Sector**
Outer Residential buildings: 288
Inner Residential buildings: 331
Outer Industrial buildings: 61
Inner Industrial buildings: 32
Outer road: 776m radius, 2438m length (~1.5mi)
Inner road: 605m radius, 1.9km length (~1.2mi)
- Midtown District & Commercial Sector**
Residential complexes: 30
Single-entity commercial buildings: 11
Commercial complexes: 10
- Uptown District & Research Sector**
Residential complexes: 20
Single-entity research buildings: 5
Research complexes: 1
- Skyline District & Robot Government Sector**
Residential apartments: 40
Government buildings: 5
- Quadrant 1 (approximates):**
Outer diameter: 800m
Height: 200m
Robot population: 4500 robots
Distance from lower Robostadt to surface: 300m

- Map colors/markers:
Green – Residential buildings
Yellow – Industrial buildings
Light gray areas – Roads and open spaces
Dark gray areas – Natural stone
Cyan – Tunnels between residential and sectors
Gold – Express elevator access tunnel
Magenta – Subway access
Purple – Old Lift access & Quadrant 4 ventilation shaft
Red – Human facilities
Dashed black arrow – Subway route (morning=clockwise)
Dashed blue arrow – Rail yard and tunnel to other quadrants
Light Blue – Storage Yards
Dark green – Storage yard security
Dark blue/Black – Nuclear Reactor Core Containment



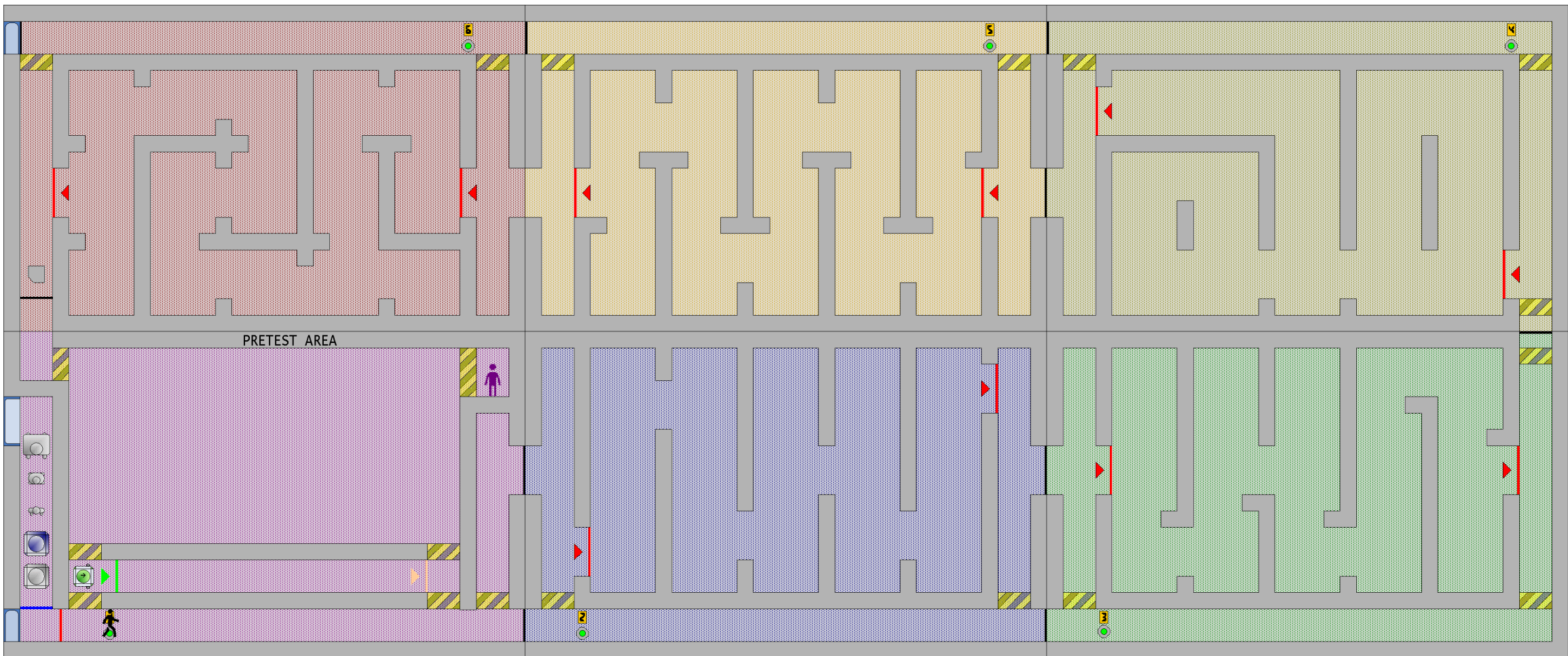
Example of screens for Lower Robostadt residential district. Most of the buildings would be locked, or would end up leading into hallways filled with locked doors, but some would be useful, containing simple puzzles or NPC dialogue. Maybe lit windows might indicate that buildings are enterable. The large archway leads from Lower Robostadt to an industrial district partitioned zone. The narrow archway leads to an express transport (elevators) and into another industrial district zone. There are barriers segmenting Lower Robostadt and preventing cross-traffic around Lower Robostadt due to the power plant shutdown. These backgrounds can probably be made as images instead of block graphics. A black sentry occasionally appears and scrolls across the screen in one direction or the other. It will ignore the player.



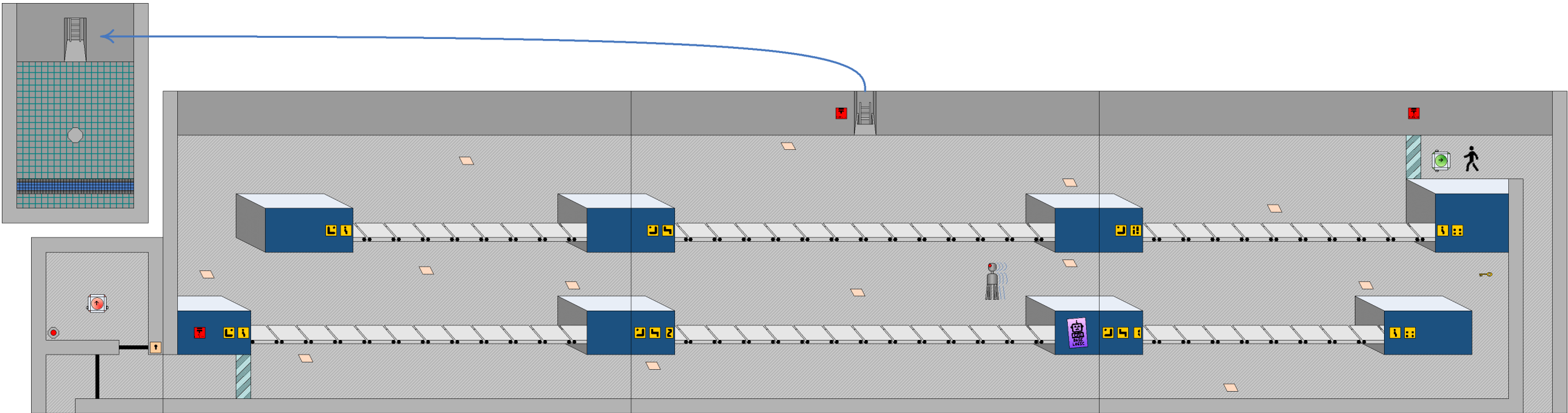
Example of screens for the Industrial district. The puzzle progression of this area consists of the Bot Proving Grounds, The Chassis Forge, and the Refining and Reclamation buildings. The other buildings will either be locked or will contain secondary puzzles or NPC dialogue. The "walls" on the sides are fire barriers that were lowered during the lockdown to reduce fire and crime risk. The Bot Proving Grounds and The Chassis Forge buildings wrap between pairs of screens to allow access once their associated puzzles have been completed.

Not Implemented

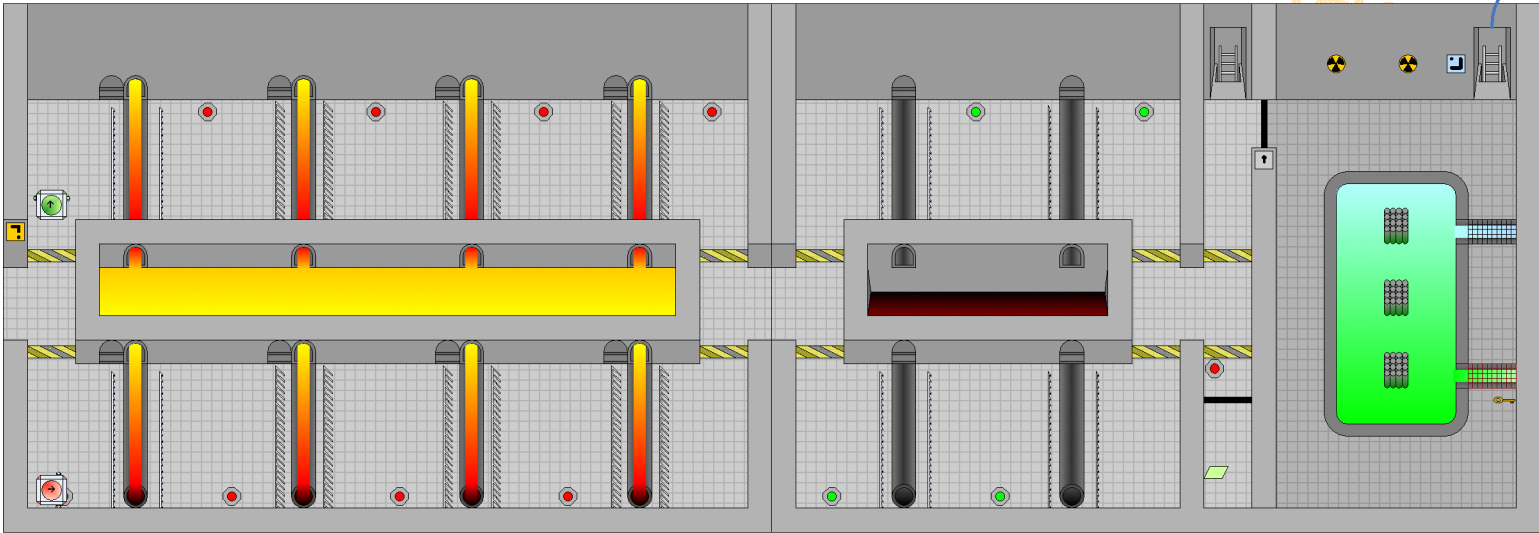
- Game level 1 steps:
1. Player starts out on Lower Robostadt with green robot.
 2. Player travels to Industrial sector zone 1 and enters Bot Proving Grounds front entrance.
 3. Player completes Bot Proving Grounds puzzle and exits through rear entrance into industrial sector zone 2.
 4. Player travels to The Chassis Forge front entrance.
 5. Player solves The Chassis Forge puzzle, rescuing the red robot.
 6. Player exits The Chassis Forge rear entrance and enters industrial sector zone 3.
 7. Player enters Refining and Reclamation through front entrance and solves puzzle, gaining an express pass to Midtown.
 8. Player exits Refining and Reclamation through rear entrance into industrial sector zone 3 and enters the Chassis Forge.
 9. Player travels back through The Chassis Forge to industrial sector zone 2 and enters the express tunnel.
 10. If player isn't ready, they can travel through express archway back to the same Lower Robostadt section they started in.
 11. If player is ready, they use the express pass to travel to Midtown district.



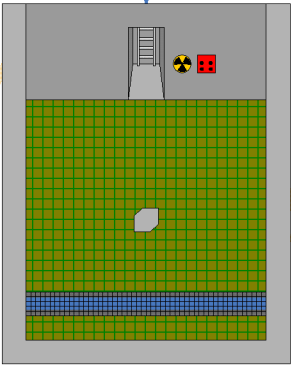
Example of the Bot Proving Grounds area. The green robot has to traverse each room in order to unlock the next room. Once the last room is unlocked, the back door access is opened up, allowing the player to access the next portion of the sealed industrial district. These would be shown as individual rooms similar to R.O. (page flipping, no scrolling, 32x20 block grid per room, 6 rooms total). The purple "evaluator" droid in the starting room will offer NPC dialogue to assist the player in understanding how these rooms operate. The first empty room is an area that the player can use to test out the robot prior to entering the maze rooms, and has a simple test at the top to unlock the start of the first maze room. For each room, the player has to position the robot at the start, then stand on the green button in that room and activate the robot. The robot needs to cross the green marker, go through the maze, then cross the orange marker while the player continuously presses the button. If successful, the black barriers disappear in that room. If the player leaves the button before the robot is finished, the orange marker will flash red indicating a failure. The player will then need to re-position the robot back before the green marker and go press the green button to restart the test for that room. The robots running a program will treat the yellow/black striped sections as walls. Robots that the user can program are shown with a top-down view. NPC Robots that the player can interact with are shown with a side profile. The scales of these robots will not necessarily be accurate.



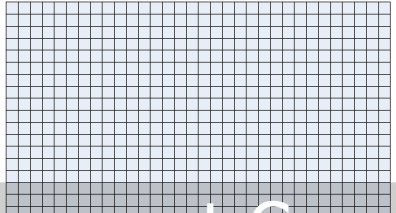
Example of the Chassis Forge puzzle. This involves the player riding inside one of the cargo robots to loop around the conveyor belts and get past the sentry in the middle while ignoring the garbage and picking up the dropped key. The key is used to unlock the security booth where the red robot, Sol, is hiding. The red button in the booth will unlock the back door and will also cause the sentry to leave. The sentry can step over the conveyor belts to remove a human if they see one outside of a robot. If the player chooses to exit a robot and take the ladder down, they will find a small room with a universal key sensor. This sensor can be placed in the gripper filter slot to restrict the gripper to only picking up keys while ignoring everything else.



Example of the molten metal storage area of the Refining & Reclamation building. One robot has to traverse the lower path while a second robot traverses the upper path. Robots have to alternately press buttons allowing the gates to drop so that the other robot can continue on their path. Due to the extremely high temperatures, the player will need to ride inside one of the two robots and cannot leave until they are back across the yellow/gray striped lines in the safe area. Ultimately the player will need to traverse this area twice, once from left to right to obtain a needed object, then a second time from right to left in order to leave. The robots will use their antennas to signal to the other robot to continue. If a robot gets permanently stuck, the player will need to restart this area. The second screen consists of an unused metal storage area that can be traversed easily due to the lowered gates and the lack of molten metal to obtain the green Midtown express pass in the closet. The second room is a radioactive storage room puzzle for later on in the game progression and requires the black robot, Lex, to take damage to obtain the key (placeholder for something else). Lex will warn that the longer it is in that room the more damage it will take, and that passing over the lower left grate will cause immediate, serious damage. If the player chooses to take some damage to use the ladder, they will obtain an EM (electromagnetic) area sensor, which might be useful for another puzzle.



- Surface
- Quadrant 1
- Quadrant 2
- Quadrant 3
- Quadrant 4



Game level 2 (Midtown/Commercial)

Areas:

1. Silicon Crucible

- 1a. Story: A new Electropower plant employee robot indicates that a memory card with their security room access code was sucked into the Silicon Crucible furnace intake and the're afraid of their code being handed over to their boss. If Ash gets the card, the employee will let her keep the card. The employee can see the card through the vent and they will be waiting on the outside of the vent for confirmation.
- 1b. Pull two levers simultaneously to allow access to crucible furnaces.
- 1c. Ride robots through the furnace ducts, pulling adjacent access levers to allow the other robot to pass.
- 1d. In the furnace chamber, press the emergency shutoff button and grab the memory card.
- 1e. Ash talks to the employee through the intake vent, confirming that they now have the card. The employee thanks her and indicates that they saw a blue compact cargo bot go into EnerPuck after it had already closed for the day.

2. Commercial Sub-level

- 2a. Story: The back door from the Silicon Crucible is locked, so Ash is forced to descend down into the tunnels which cross under the street. This path will lead up into a utility closet in one of the the BotLogic Data Mining rooms.
- 2b. One robot is sent through the maze, but encounters three one-way vent obstacles.
- 2c. A sentry will interfere with any human trying to pass through the central maze area.
- 2d. A second robot needs to cross the upper hallway, triggering the vent reversal controls at the appropriate times to allow the lower robot to pass.
- 2e. Once the lower robot reaches the end of the maze, it presses a button opening the locked up-ladder.

3. BotLogic Data Mining

- 3a. Story: The bots figure that BotLogic is a simple office setting and they should just be able to walk from the front door to the back door through one of the unlocked interface rooms, unfortunately, they are caught by one of the BotLogic repair technicians who locked them in and threatened to call the sector police. The technician has been tasked to repair an interface that was receiving interface errors, but they indicated that they accidentally deleted the email message telling them which interface was faulty. They agree to release Ash and the bots if she helps determine which interface is faulty.
- 3b. The technician indicated that test data was being sent to each interface. We need a program that will read the data words from the interface, split each 32-bit word into four bytes, XOR the bytes together, then send the single-byte result back to the interface. Once we read a word that is zero, we are finished with that interface and the robot can move to the next interface. The interface has a display that will light up with green or red lines indicating success or failure. Once we finish all 15 terminals, the technician will unlock the door. We are expecting errors for exactly one interface, but if we get incorrect results for more than one interface, we will need to re-run tests on those interfaces until only one is showing an error. We can also have both robots each helping out with half of the interfaces to speed up the process.

4. EnerPuck

- 4a. Story: Ash exited the BotLogic building and headed for the EnerPuck building to find the blue robot, Lev.
- 4b. Ash entered one of the EnerPuck assembly line areas and needs to get past welding lasers and a gray sentry. Ash needs to program one robot to use an electromagnetic area sensor and electromagnetic contact sensor to determine when to safely move or stop. That robot needs to signal to a second robot when to start and stop moving. Once they get to the far east wall, they will move upward and simultaneously press buttons unlocking the exit doors. The blue robot, Lev will be waiting at the end of this puzzle.

5. Electropower plant

- 5a. Ash and the bots have now rescued the third compact cargo bot and exited the EnerPuck building. The only other exit from this area is to enter one of the vents into the Electropower plant and travel to the northeast exit.
- Possible puzzles:

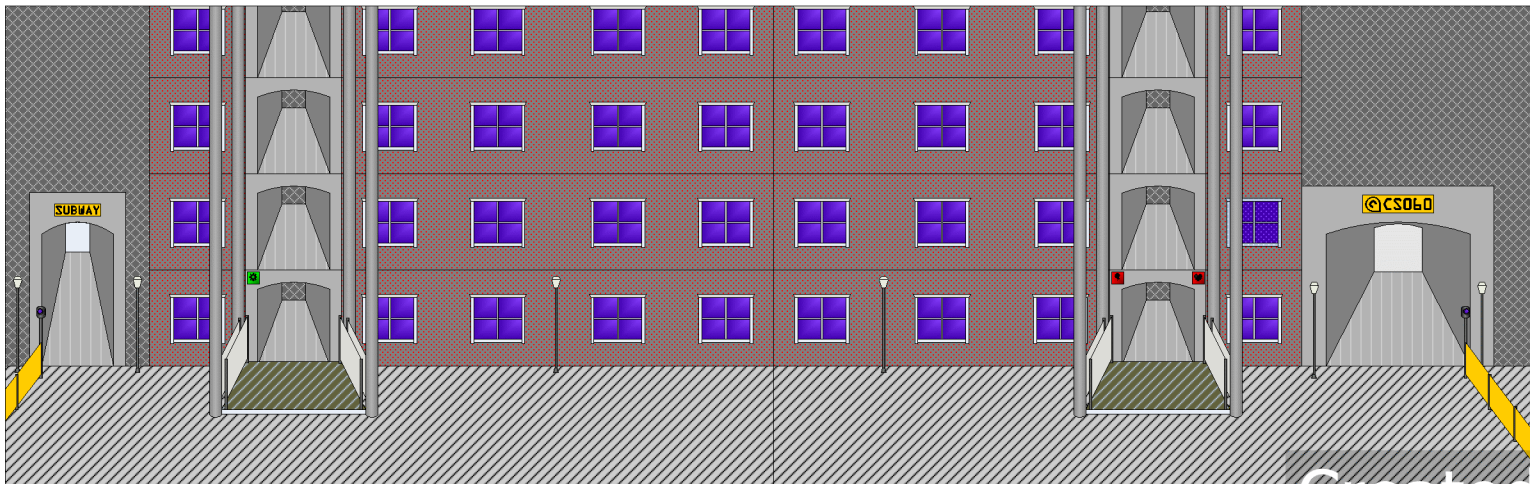
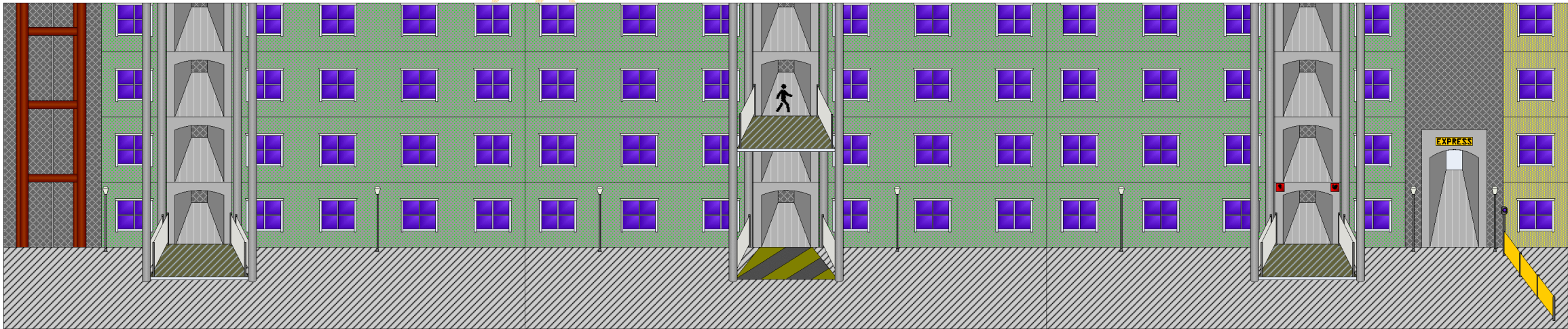
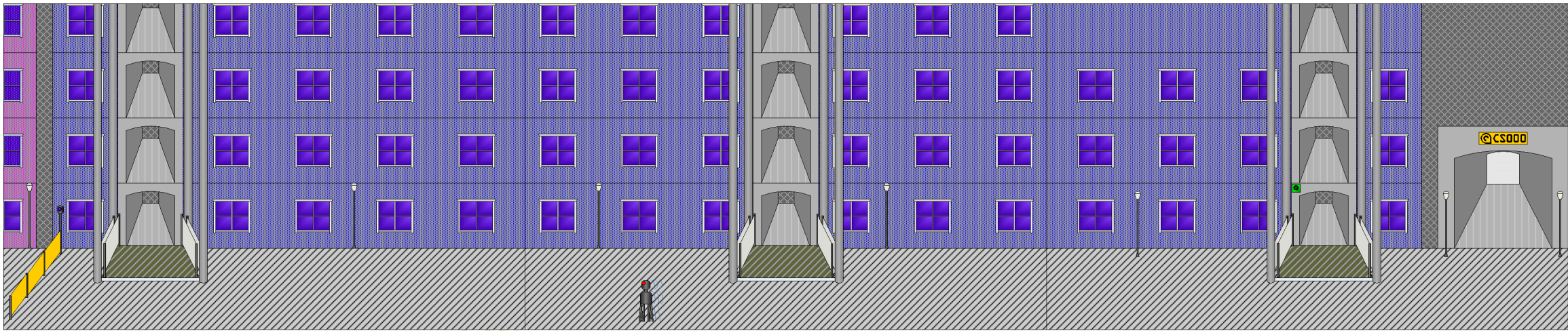
A. Two robots unlock a door to let the third robot into a security room (one transmits a code to an interface while the second pulls a lever to open the door). While the third robot locates the security interface, one of the first two robots move down to a locked security door with an interface, while the second robot has to keep the security room door open by keeping the lever pulled. The robot in the security room obtains a temporary key and transmits it to the robot at the security door, who then unlocks the security door. The security key is only valid for 60 seconds. The security room will have multiple consoles in it, but only one will be working properly, so the robot in that room will need to check each one until it finds the working one. Note: this puzzle relies on having a security room access code, which was obtained from the Silicon Crucible puzzle. That memory card will effectively be read-only until after this puzzle is completed.

5. Electropower plant

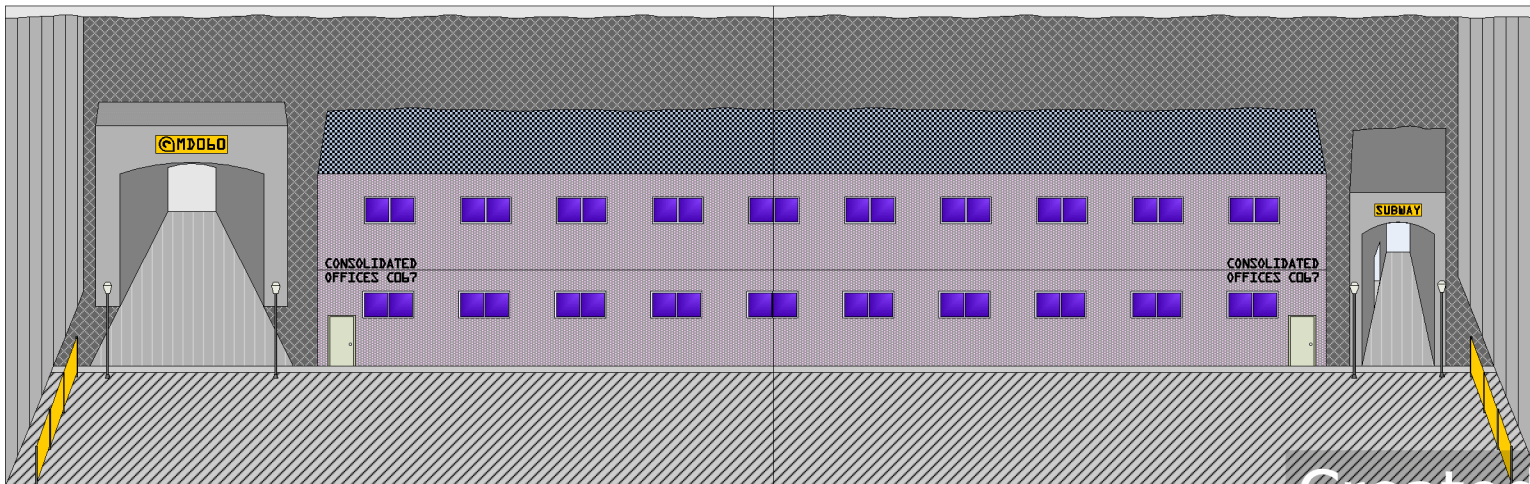
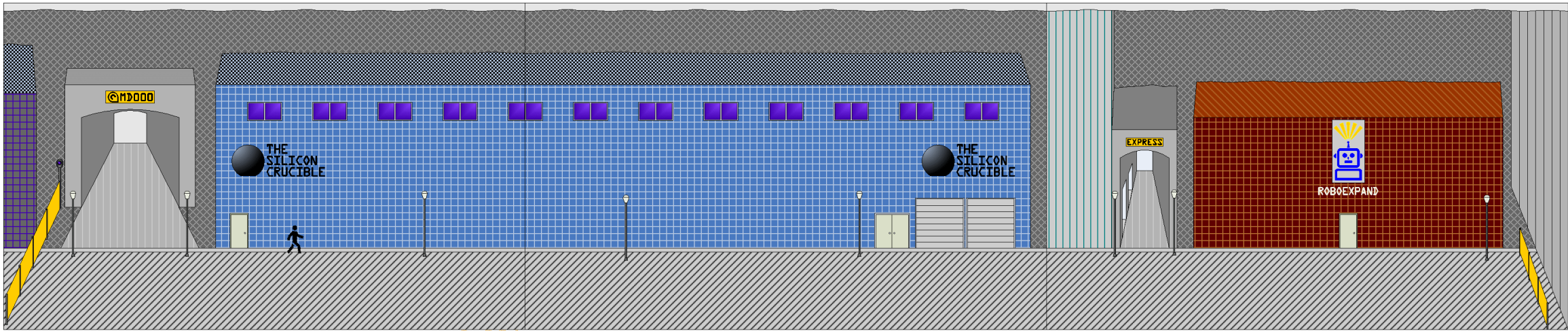
- 5a. Ash and the bots have now rescued the third compact cargo bot and exited the EnerPuck building. The only other exit from this area is to enter one of the vents into the Electropower plant and travel to the northeast exit.

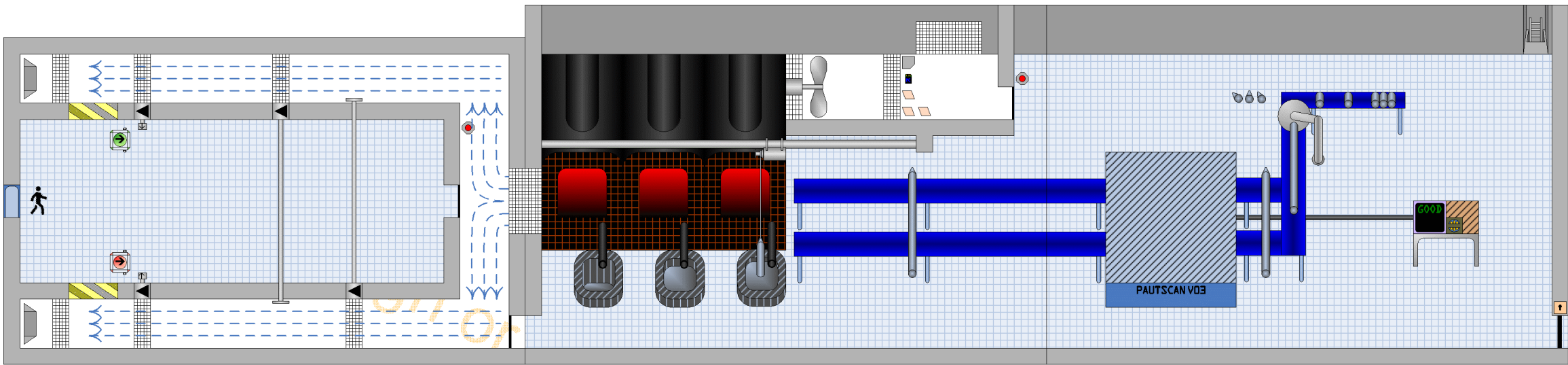
5b. puzzles

1. Security office.
- 1a. Two robots unlock a door to let the first robot into a security room (one transmits a code* to an interface while the second pulls a lever to open the door).
- 1b. While the first robot locates the security interface, the third robot moves down to a locked security door with an interface, while the second robot has to keep the security room door open by keeping the lever pulled.
- 1c. The robot in the security room checks each interface for the correct key. The security key is only valid for 60 seconds. The security room will have multiple consoles in it, but only one will be issuing valid keys, so the robot in that room will need to check each one until it finds the working one. To verify the key, the 32-bit code received via the interface will need to be broken up into four bytes, then each byte needs to be exclusive-OR'ed (XOR) with each other and the result needs to be a byte with all ones (0xFF or 255). If any other result is obtained, this is not the correct security terminal and the next one will need to be checked.
- 1d. Once the robot obtains the correct key, they transmit it via radio to the robot at the lower security door, who then unlocks the door. The key will only be valid for 60 seconds, so there will not be enough time for the robot in the security room to exit the room and move down to the lower security door. Once the lower security door is unlocked, the two blue beams will also disappear.
- 1e. Notes: * this puzzle relies on having a security room access code, which was obtained from the Silicon Crucible puzzle. That memory card will effectively be read-only until after this puzzle is completed.
2. Distorted maze.
- 2a. Story: When the robots first arrive, a worker bot tells them about how the hallways have been blocked off and diverted into a strange maze, and the dangers of robots and humans being exposed to the high-frequency radio waves. It indicated that it had purchased an EM shield, but that its auxiliary slot is broken, and explains how we can use that to traverse the maze.
- 2b. Consists of random directional EM currents strong enough that Ash needs to ride inside of a robot.
- 2c. The robot will need the EM shield to block the EM in the direction it is flowing from (NC/EC/WC/SC flags), and the EM directional sensor to detect the direction (NS/ES/WS/SS flags). When the EM shield is up, the thruster and wall bumper on the active EM shield side(s) will not be functional.
- 2d. The robot will also need to reverse its hover pad (HR flag) to lock itself in place when the EM current arrives, otherwise its thrusters will fire randomly and push the robot around. The EM area sensor (AS flag) can be used to detect that.
3. Quick charge battery
- 3a. Once the robots have exited the maze, they discover locked doors that are disconnected from power, however they have battery slots for emergency power. The room has a battery charger and some batteries, but all appear to be faulty batteries that can barely hold a charge. The door locks have capacitors with enough power to pull the doors open, however they need to detect a valid battery in the slot before they will trigger.
- 3b. Ash must use a bot to drag the batteries over to the charger and see if the battery charger will accept any of them (flashing red light = bad battery, solid red light = charging, green light = charged). The bots can pull, push, and drag batteries around with their thrusters and grippers.
- 3c. One battery will be suitable to solve this issue. Ash's program must be optimized to precisely move the battery, because the battery will start losing charge as soon as it is removed from the charger.
4. Bonus crate puzzle
- 4a. Ash and the bots are standing in one of the delivery areas of the power plant, and they spot a sensor module of some sort behind some of the crates. Ash can program a bot to move the boxes around to get the sensor module. The module will make one of the upcoming puzzles a bit easier to program.



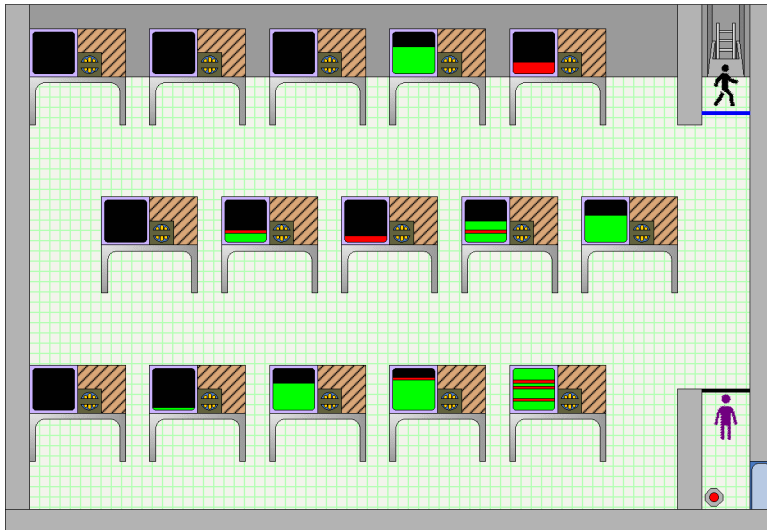
Example of screens for a small section of the Midtown residential district of Robostadt. This area consists of large residential buildings with front-access lifts instead of internal elevators or stairs, and a long hallway on each level going end-to-end down the center of the building, with fire-escape slides on the back of the buildings. The large archway leads from Midtown to a currently-partitioned commercial district zone. The narrow archway leads to an express transport (subway and elevators) and into the commercial district. There are barriers segmenting Midtown due to the power plant shutdown, but are not as numerous as Lower Robostadt.





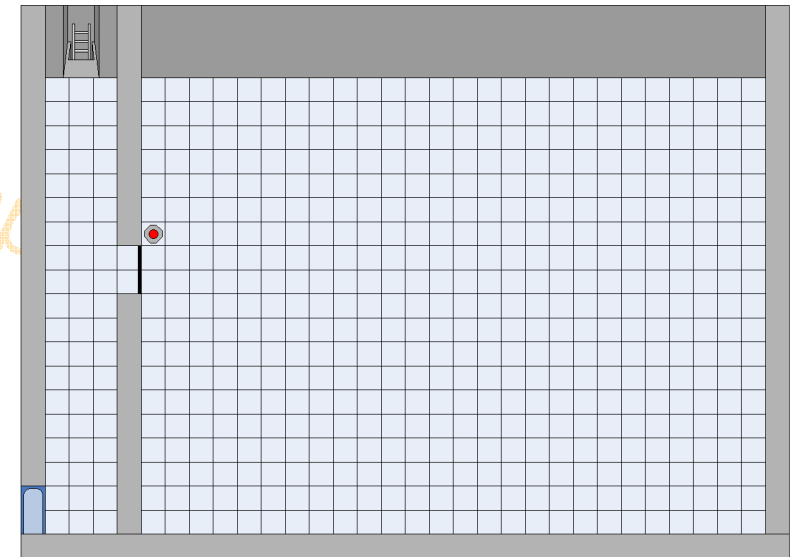
Level 2a (Silicon Crucible) Puzzle steps:

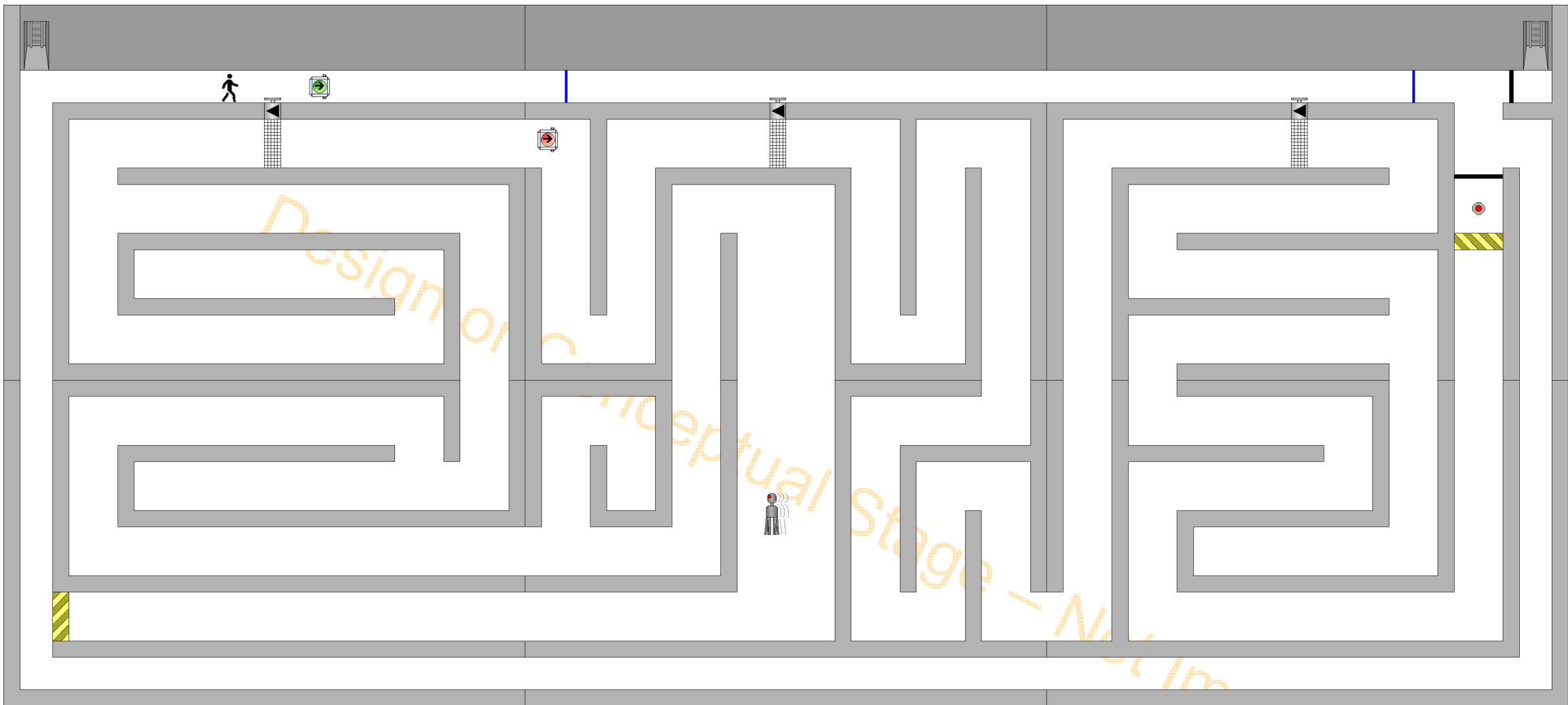
1. One robot is positioned to grab the upper knob and pull downward, the other is positioned to grab the lower knob and pull upward. They must be pulled simultaneously.
2. Once knobs are fully extended, robots release the knobs, which slowly slide back closed over about 10 seconds.
3. Robots move through the yellow/black striped zones into exhaust air current and move rightward through their open grilles.
4. The upper robot moves up against the second closed grille. The lower robot toggles its gripper on and off, which reverses the one-way grille for about 5 seconds to allow the other robot to move rightward through it. Note: slideable or reversible grilles like this will not trigger wall bumper sensors.
5. The process is repeated, but with the upper robot toggling the lower robots one-way grille.
6. After both robots are through the grilles, they move rightward, and then toward the center.
7. One robot pushes the button, unlocking the black doors and shutting off the exhaust fan, allowing the player to continue.
8. If the robots get stuck, the player just has to halt the robots using the remote and the exhaust currents will push the robots back to the start where the player can retrieve them.



Level 2c (BotLogic Data Mining) Puzzle steps:

1. A robot should be programmed to move from interface to interface, performing I/O actions on each interface. The task can also be split between both robots.
2. Once a robot connects to an interface, a series of random 32-bit values are read via the interface register. For each 32-bit word read, split it into four bytes and exclusive-or (XOR) the bytes together. For example, if the word read was 0xEF9A3B4C, the four bytes would be 0xEF, 0x9A, 0x3B, and 0x4C, and the XOR result would be 0x02. It doesn't matter which order the XOR operations are performed in, the result should always be the same.
3. Write the single-byte result to the interface.
4. Continue this process with each additional 32-bit word until a zero value is read. If the robot accidentally connects to an already successfully-finished interface, the first value it will read will be a zero. If the interface had received an error, then reconnecting to it will restart the testing process for that interface.
5. As these value are written to the interface, the display next to the interface will slowly fill up with green or red lines.
6. Once all interface tasks have been successfully completed (with exactly one showing red errors and all others being solid green), Ash and the bots will be allowed to leave.





Level 2b (Commercial sector sub-level) Puzzle steps:

1. One robot is sent through the maze, but will encounter three vent obstacles blocking its way. A gray sentry is present in the middle of the maze to stop the player from manually walking through the maze, and the blue beams prevent the human from crossing the top hallway. To overcome the obstacles, a second robot needs to move across the top hallway above the maze and open the one-way vents at the appropriate time to let the lower robot pass. Once the lower robot has passed all three vents, it should be able to press the red button and open the black doorways. The unlocked up-ladder leads to one of the Botlogic Data Mining rooms.

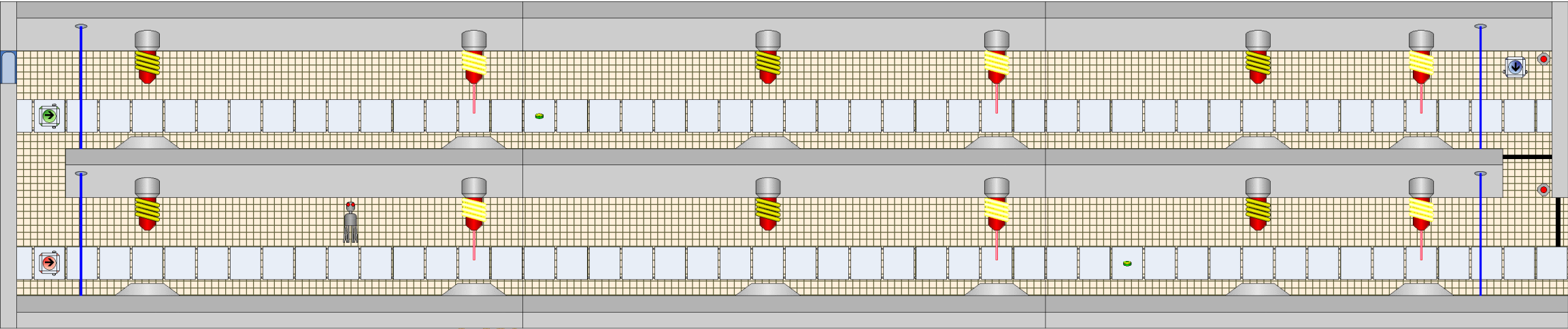
2. Hints:

The robot area sensor obtained in the Silicon Forge intake vent could be helpful in signalling when the upper robot should grab or release each vent switch before moving onto the next one.

The rotational and pull-style one-way vents are not detectable by the wall bumper sensors, but will still block the robots from moving.

If the robot in the maze is unable to push the button (soft lock), a game save will need to be reloaded to reset the level.

Level makeup: Six screens, with sizes between 32x20 and 32x22, with scrolling disabled.



Level 2d (EnePuck building):

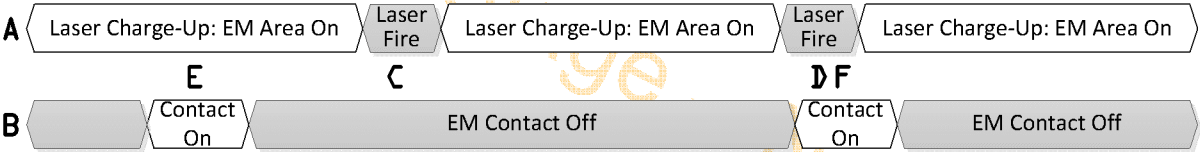
- 1. A robot has to move through a hazardous area where energy cells are being welded. The welding beams will fire sequentially, but an electromagnetic field will form just before the laser fires. Ash will need to ride inside a robot. The robots will need to move in unison, and both robots will need to stop precisely when one robot detects a safe point to stop, and start back up when the threat is gone. Both robots need to press switches simultaneously to shut off the assembly line and rescue the blue robot, Mel. If the robots move through the lasers, they will be damaged. The conveyor belts are powered off and will not affect robot movement.
- 2. The electromagnetic area sensor will be triggered when the lasers are heating up but not yet firing. A second sensor, an electromagnetic contact sensor will indicate when the robot is in the line of fire of the laser path. If the robot detects the EM contact, it needs to continue forward until the sensor deactivates. The area EM sensor will deactivate when the laser fires, but will reactivate right after the laser has finished.
- 3. Since there is only one set of sensors, the robot with the sensors will need to use its radio transmitter to send a stop/start signal to the other robot.

Simplistic program actions for sensing/guiding robot:

- 1. Is electromagnetic area sensor and/or contact sensor engaged? Enable antenna signal and start eastward movement.
- 2. Is electromagnetic contact sensor and area sensor both disengaged? Disable antenna and stop movement.
- 3. Is right wall bumper sensing a wall? Release gripper and engage south thruster to push button.

Simplistic program actions for guided robot:

- 1. Is radio signal active? Start eastward movement.
- 2. Is radio signal inactive? Stop movement.
- 3. Is right wall bumper sensing a wall? Release gripper and engage south thruster to push button.

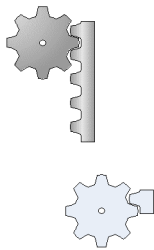


- A: Electromagnetic Area Sensor state, light=on, dark=off
- B: Electromagnetic Contact Sensor state.
- C: Firing of leftmost (or rightmost) laser pair on each page.
- D: Firing of rightmost (or leftmost) laser pair on each page.
- E: Robot successfully passing under inactive laser.
- F: Robot(s) being damaged by laser in this state, assuming that the laser firing is the one the robot is passing under.



Room size: 9 rooms, each 31x21

Story lore:
When the robots are shrunk down and placed in the carrier, and the carrier is inside another robot, that robot can transmit their video and other sensory inputs to the shrunk robots so that they can see what is going on around them.
In this instance, in the middle of the maze, the robots saw what appeared to be two black sentries in a booth acting like they were having a conversation, even though those robots shouldn't have any capability for that. Further more, one of the sentries appeared to have a red light at the base of its neck, prompting Sol to tell the story about the "black wraith" sentry. The robots also expressed bewilderment as to what this maze is and what the machines are that are causing the interference.



Example advanced puzzle: 

You have to get a specific robot program which allows it to traverse a restricted area. The program data is split randomly into four separate pages available for download from four interfaces in the room. Each chunk of data is a 16x16 program page with only a random ¼ of the data spread across the page. Combining the four partial program pages together properly will give you the complete program. The partial pages might be rotated or flipped horizontally or vertically in some random combination that changes (along with the contents of the pages) each time you try this puzzle. Once enough partial pages are downloaded, you need to use a program (or subroutine) that can determine how the pages need to be adjusted to be properly combined to build the complete program. The robot can then run the completed program at the appropriate spot to get through the restricted area. To make this puzzle more challenging, you have a limited time to complete the whole puzzle and run the resulting program. This time limit requires that you simultaneously combine the downloaded segments while you are still moving the robot around obtaining the other segments, which means the program needs to autonomously cycle through the various flip/rotation permutations to find the correct orientation. The final combined program also has to be rotated correctly for the proper 0,0 starting point to be used.

- Hints:
- * Each of the four partial pages will have one corner filled and the other three corners empty. This may provide a shortcut that can quickly trim away ¼ of the checks.
 - * One page always contains the starting program corner cell as one of its corners. This page may be rotated, but will never be horizontally or vertically flipped. The other three will always be flipped either horizontally or vertically and might also be rotated. This should give you the ability to know how to orient the overall combined program to start properly and which corner is the program start.
 - * You can tell if pages are properly aligned with each other if there is no common data point in the 16x16 grid on the pages where non-zero data conflicts between the pages. Once the pages are combined, there should be no empty cells.

Example sort puzzle:

Backstory:
At some point in the game, you realize that all of the first generation robots of Robostadt have had a certain portion of their memories mass-erased, but you ultimately find someone that was undergoing major CPU module repair when the erasure was taking place. After they were again operational, they still had the 'unauthorized' memories, and at the time, the concept of pushing memories to offline storage was not yet a reality. So that they wouldn't be found out by the mainframes and punished for still having those memories, they unlinked them from their own memory structure. They are aware that the memories are still there, but they can't access them. Their CPU module repair was not able to fix some burned-out high bits in their ALU, so they are not able to sort the 32-bit memory indexes themselves if they wanted to recover the memories.

Puzzle:
They are willing to send (via interface) the random 256 unlinked indexes to one of your 'bots, and you have to write a sort utility to sort the 32-bit numbers in ascending numeric order and then send the data back out the interface to them.

Reward:
Once they receive the numbers in the proper order, their low-level memory manager can properly scan for and relink the memories back into their memory database. They can then dump the memories one by one to a high-capacity memory card you obtained earlier and delete them from their own internal memory.

- Hints:
- * Since the robots do not have enough register space, the numbers will need to be written directly to program pages or memory cards as raw data.
 - * Since there is no speed requirement for reading the numbers from the interface before sorting, that can possibly be incorporated in the sort routine itself, so long as it doesn't slow down the overall job.
 - * Any suitable numeric sort routine can be used to sort the numbers, so long as its capable of being programmed with the available opcodes. Some high-recursion sort routines may not run properly.

Example advanced networking puzzle:

Backstory:
You are trying to look up a security door code from a secure storage facility. You will use the data token you received from the previous puzzle. The first part of the data token (bits 0-2) is a 3-bit network address which determines the storage vault you need to talk to. The second part (bits 3-15) is the 13-bit lookup key, which determines if you are authorized to get the information. If the drone ends up going to the wrong storage vault, the data token will be invalidated and you will have to repeat the previous puzzle over again to get a new data token. If something goes wrong, the player can press the alarm button on the wall and the sentries will remove any robots in the network nodes. Every time the alarm button is pressed a 5th time, the data token will be invalidated and the player will have to repeat the previous puzzle. The player and the regular robots are too large to enter the network, so one robot will need to be upgraded to a mini-drone to complete this puzzle.

- Puzzle:
Access to the storage vault is via an omega network that utilizes an XOR tag address. To traverse the omega network:
1. Send the drone through the entrance. They will be teleported into one of 8 small hallways, each with an interface.
 2. Connect to the interface and read the source address from the interface.
 3. XOR the source address with the 3-bit destination address to get the XOR tag.
 4. Move the robot east. Once it hits the east exit, it will be teleported into the first network node.
 5. In the first node, the robot will either be at the upper entrance or the lower one. If bit 2 of the XOR tag is a zero, move the robot straight east to the exit. If bit 2 is a one, move east and north or south to the other exit (diagonal from the entrance) and go through that exit.
 6. Repeat step 5 for the second node, but use bit 1.
 7. Repeat step 5 for the third node, but use bit 0.
 8. Move east to the interface and connect. At the interface, send the data token and retrieve the security door code. If one of the previous steps were not followed correctly, then the alarm will sound and the sentries will remove the drone and the previous puzzle (and this one) will need to be repeated.
 9. Once the security code is obtained, reverse the steps (and direction) to move back to the original entryway.

- Hints:
- * As with other standard omega network designs, the player can also use destination tag addressing where you only use the destination address (without the XOR), and a zero bit = always uses the top exit, and a one bit = always use the bottom exit. To exit the network, the source address is used.
 - * With the drone in a narrow passage, both the north and south (or east and west) wall sensors will be in contact, and the drone will be forced to the center. In these type of passages, a wall sensor signal going low might be a more important detection event than a wall sensor signal going high.

Example challenging environmental puzzle:

Location: Commercial District, Electropower Plant, Central power chamber, secondary puzzle

Description:
Extreme electromagnetic environment will prevent non-shielded robots, thrusters, hover pads, and shrink rays from operating. Even the black robots cargo compartment isn't quite sealed enough for this area. Without operational hover pads, the robots will be too heavy for the human to move. All four robots and the player will need to cross the area as quickly as possible.

Puzzle:
The black robot will need to be upgraded to use a tank tread attachment* instead of thrusters, and then will need to be programmed to grab and pull each robot individually through a narrow path across the hazardous environment. The silver robots will need to be placed on rolling carts, then shut down and powered off during the process. The human can go across the area, but must be very quick or they will get injured. The backpack will also auto-power off in the area.

Risks:
The black robot will receive damage based on their exposure time in the environment; they may need a repair token after this puzzle. If the stray too far off of the narrow path crossing the chamber, they will trigger a motion alarm which will alert the sentries. If the human gets injured enough by microwave radiation, they may lose the ability to carry robots or the backpack until they get healed up via some rest and relaxation doing tutorials, participating in a robotour, or completing some mundane jobs in a residential district.

* Tank tread attachments and other upgrades can be ordered through upgrade shops in the residential areas, although not every upgrade is available on every level. The upgrade shops have a back room where upgrades can be tested.

Game Concept: CPU² Robots (Continued) – Wall Crawler Program

[illegible]

This is my first attempt at a CPU² program. This should theoretically be a right-wall crawler program, but I currently have no way to test it. The first process creates five other processes: four processes to handle regular wall movement, and one to handle outside corner walls. Register 9 is converted to a group register and is used to keep track of the last activated thruster. Some of the delay values might have to be adjusted or moved around in the code, and this program likely will not be able to handle very erratic wall conditions without further refinement.

This will no longer function because of changes in the opcode table.

[illegible]

This might be a better design for a right-wall crawler program. The cyan process monitors the bumper statuses in the flag register, then uses the shared 'GO' flag to signal the other processes as to whether they should handle regular wall movement or outside corner mitigation. Each of the other four processes examine that shared flag to determine which mode they should be in. They then either perform their regular wall movement (and update the shared register '9' if they engage their thruster), or in the alternate mode, they examine the shared register to see if they are the last thruster to fire and if so, they turn on a pair of thrusters to correct for the outside corner. Once any of the wall bumpers make contact again, then the cyan process switches back to the regular mode using the shared flag. The delay opcodes have been left out of this and can be added back in if necessary. This doesn't use any 'redo' conditionals, so it shouldn't be prone to any potential process lock-ups.

This will no longer function because of changes in the opcode table.

	TERM	FORK, RR	NOP	LDB 0x01	RREG, L	VIREG, 0	TERM	THR,W, OFF	CSF,NB, RR	THR,W, ON	RREG, 0		
			NOP	LRET	THR,W, ON	VIREG, B2	THR,A, OFF	LRET	NOP	LRET	VIREG, F		
			NOP	THR,E, ON	CST,G1, RR	THR,S, ON	LOOP	CST,G2, RL	CST,A, RL	CST,G1, RL	CSF,NB, RR	THR,E, OFF	LRET
			NOP		LRET	CSF,D, RL	XOR	VIREG, B1	RREG, F	THR,E, ON	RREG, 0	VIREG, F	
		FORK, RR	NOP	LDB 0x01	RREG, L	VIREG, 0	TERM	THR,N, OFF	CSFEB, RR	THR,N, ON	RREG, 0		
			NOP	LRET	THR,N, ON	VIREG, B2	THR,A, OFF	LRET	NOP	LRET	VIREG, F		
CLR G8			NOP	THR,S, ON	CST,G1, RR	THR,W, ON	LOOP	CST,G2, RL	CST,A, RL	CST,G1, RL	CSFEB, RR	THR,S, OFF	LRET
CSTD, RL	SET G0		NOP		LRET	CSF,D, RL	XOR	VIREG, B1	RREG, F	THR,S, ON	RREG, 0	VIREG, F	
AND	VIREG, B1	FORK, RR	NOP	LDB 0x01	RREG, L	VIREG, 0	TERM	THR,E, OFF	CSF,SB, RR	THR,E, ON	RREG, 0		
TERM	RREG, FS1		NOP	LRET	THR,E, ON	VIREG, B2	THR,A, OFF	LRET	NOP	LRET	VIREG, F		
THRA, OFF	CSF,G2, RL		NOP	THR,W, ON	CST,G1, RR	THR,N, ON	LOOP	CST,G2, RL	CST,A, RL	CST,G1, RL	CSF,SB, RR	THR,W, OFF	LRET
	LOOP		NOP		LRET	CSF,D, RL	XOR	VIREG, B1	RREG, F	THR,W, ON	RREG, 0	VIREG, F	
VIREG, F	VIREG, B2	FORK, RR	NOP	LDB 0x01	RREG, L	VIREG, 0	TERM	THR,S, OFF	CSF,MB, RR	THR,S, ON	RREG, 0		
RREG, L	LDB 0x0F	FORK, RL	LRET	THR,S, ON		VIREG, B2	THR,A, OFF	LRET	NOP	LRET	VIREG, F		
TERM	PRET	CLRL	THR,N, ON	CST,G1, RR	THR,E, ON	LOOP	CST,G2, RL	CST,A, RL	CST,G1, RL	CSF,MB, RR	THR,N, OFF	LRET	
START	FORK, RL	ACOFF		LRET	CSF,D, RL	XOR	VIREG, B1	RREG, F	THR,N, ON	RREG, 0	VIREG, F		

This is an expansion of the middle wall crawler design. This can switch between right and left wall crawling (using the 'G1' flag), and terminating the wall crawler processes (using the 'G2' flag). The 'G0' flag is still used for the movement mode, and shared register 'F' is now used for last active thruster ID storage. This also uses the updated conditional flag functionality, substantially shrinking its size.

Left Wall Crawler:
West bumper activates South thruster #1
North bumper activates West thruster #2
East bumper activates North thruster #3
South bumper activates East thruster #4
North Thruster #1 correction uses South & West thrusters
West Thruster #2 correction uses East & South thrusters
South Thruster #3 correction uses North & East thrusters
East Thruster #4 correction uses West & North thrusters

Right Wall Crawler:
West bumper activates North thruster #1
South bumper activates West thruster #2
East bumper activates South thruster #3
North bumper activates East thruster #4
North Thruster #1 correction uses South & East thrusters
West Thruster #2 correction uses East & North thrusters
South Thruster #3 correction uses North & West thrusters
East Thruster #4 correction uses West & South thrusters

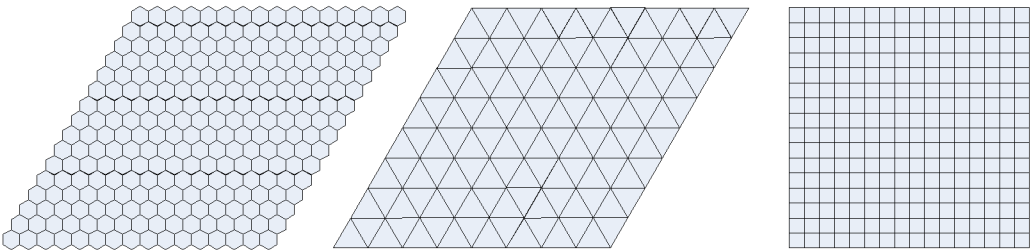
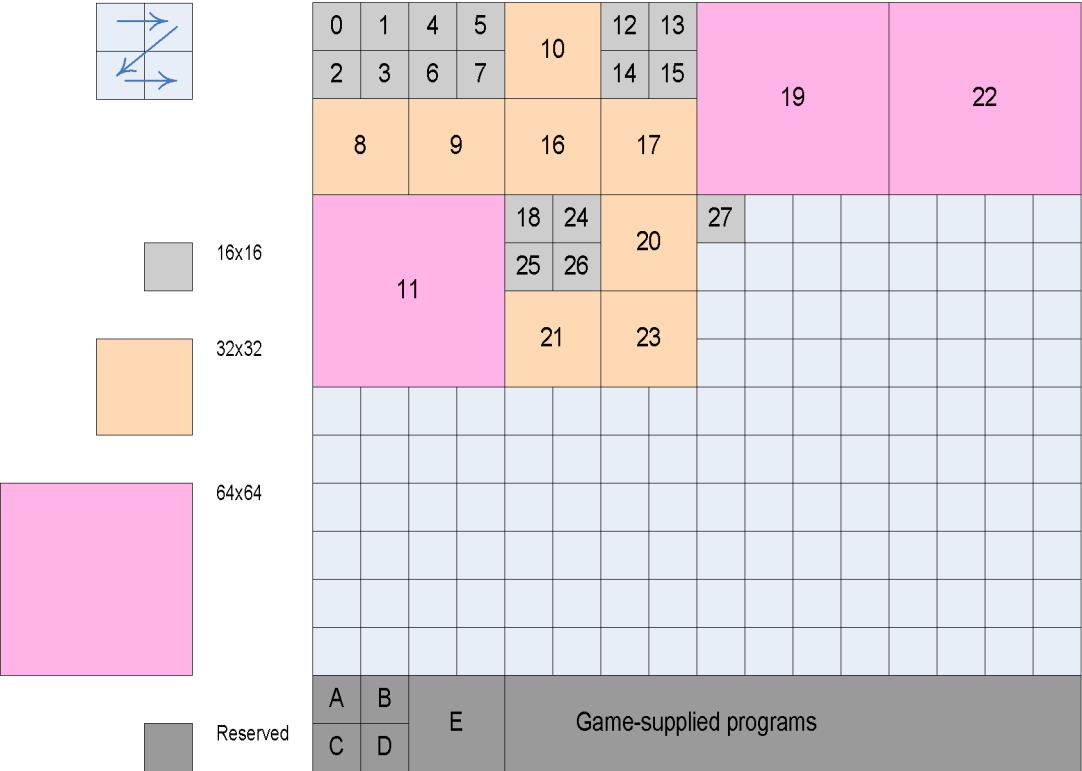
CST, **	Flag is Zero/False
Flag is One/True	CSF, **
▷ Regular program execution ▶ True conditional direction change	
RL	RR

Inst. Word:0x0000F0E

Inst. Word: 0x7200007D
Real Opcode: COND,B
Conditional: CST.ROT.RL

[illegible]

Robot programming area recursive page fill example:



Final Stage – Not Implemented